MARKET SCANNER BY DEV

Complete Conversation Record

F&O Master Fix Stream · P0.4 Skip-Reason Diagnostics · Money-Path Instrumentation

Live-Market Incident & Recovery · F&O Completion Mission · Pre/Post Briefing Feature

Period: 15–17 July 2026

Owner: Devendra · Hrishi Associates (marketscannerbydev.in)

Advisor: Claude (Anthropic) · Coder: build agent (reports relayed by owner)

Contents: Parts 1–6 — full turn-by-turn record with embedded screenshots

Appendices A–D — the four source documents reproduced verbatim in full

F&O Master Fix Stream · P0.4 Diagnostics · Money-Path Instrumentation · Completion Mission

Period covered: 15–17 July 2026 · Owner: Devendra (Hrishi Associates) · Advisor: Claude

Note: Every exchange in the conversation is recorded in order. Speaker turns are complete in substance; formatting is normalized for print. Large source documents are reproduced in full as Appendices A–D. All nine uploaded screenshots are embedded at the point of upload.

PART 1 — SCOPING THE MASTER FIX & LAUNCHING P0.4

DEVENDRA — Phased plan for FNO_SWING_MASTER_FIX

Presented the full doc sliced into ~15–25 dev-days: P0 Financial truth (P0.1 kill false "Market is closed"; P0.2 /api/options/signals schema drift + degraded-state contract; P0.3 pre-open Kite readiness gate; P0.4 skip-reason diagnostics with full candidate→open trace), P1 Reliability+honesty (data infra, canonical bias, paper execution with 12 exits, swing state machine, scanner), P2 Reports+strategic (report suites, backtest provenance labels, calibration). Asked: (1) which slice first — recommended P0.1+P0.2 together; (2) R1-tail baseline fixture folded or separate; (3) any escalation, e.g. if F&O zero-opens is live in prod, P0.4 jumps.

CLAUDE

Answered (a) P0.1+P0.2 — but with a 15-minute evidence pre-step: run a read-only prod query counting F&O paper-trade opens vs candidates over the last 5–10 sessions; if candidates flow but opens are ~zero, P0.4 jumps to the front (evidence-first rule). Explicitly rejected option (d) "all P0 in one push" — it removes per-slice reconciliation checkpoints, the only defense against the coder's documented adjacent-drift pattern. R1-tail: keep separate (needs a live Monday, zero code; coupling creates fake dependency). Escalation: conditional on the pre-step only. Two technical flags: contractInstrumentToken harmonization must be additive-only via ALTER TABLE ADD COLUMN IF NOT EXISTS, harmonize in the code layer, never rename prod columns; P0.1's test list should add NSE special sessions (muhurat, shortened expiry days). Provided a paste-ready reply enforcing one-slice-one-PR, literal test evidence, no adjacent actions.

DEVENDRA — "can you access my github repository if given a link"

CLAUDE

Public repos: yes — fetch specific pages, or clone the whole repo in the sandbox (github.com allowed) for grep/tests/audits. Private repos: no GitHub login; options are a fine-grained read-only PAT in the clone URL (treat as exposed, scope minimally, revoke after) or the zero-risk route: upload the repo as a ZIP. Everything is read/analyze — Claude can produce patches and diffs but cannot push, open PRs, or write back. Suggested pattern: ZIP/token → clone → audit → patch files + literal test evidence.

DEVENDRA — P0.4 slice directive

Start P0.4 next iteration: persist full candidate→signal→tier→gate→contract→sizing→execution→open trace (fno_signal_reasoning is 80% there), read-only /audit ops panel with funnel counts + top rejection reasons, investigate WHY 100% of EMITTED signals end at BASELINE (suspects: veto→INFO_ONLY over-firing, confluence threshold, POST_CLAMP_RR clamp), additive schema only, replay last 6 BASELINE emissions proving each had a specific gate reason.

CLAUDE

Restructured the slice with investigation BEFORE panel-building. Step 0 (read-only, ~30 min): query the last 6 BASELINE emissions from fno_signal_reasoning and paste raw rows — the silence point identifies the instrumentation target; do not build the panel against the wrong stage. Step 1: instrument the identified demotion gate with gate name, threshold, actual value, verdict — reasons from an enum taxonomy extending the existing 27-reason set, never free text. Step 2: append execution-stage rows. Step 3: /audit panel strictly read-only — no retry/override/tuning controls. Step 4 acceptance: (a) replay of 6 BASELINE emissions each with a specific gate reason, zero nulls; (b) literal pass lines from pnpm --filter @workspace/api-server run test in Replit Shell with --pool=threads; (c) funnel counts that sum (candidates = rejections + opens). Bright line: this iteration DIAGNOSES the 100%-BASELINE issue; even if the cause looks obvious (e.g. threshold 0.8 vs scores 0.75), report and stop — tuning is a separate owner decision.

DEVENDRA — Step 0 raw evidence (iteration 16) summary

All 6 rows structurally identical: setup_key=BASELINE, tier=BASELINE, decision=EMITTED, reason_code=EMITTED (born baseline, not demoted), confluence_score NULL (bypasses confluence engine), option fields NULL, data_quality=LIVE_KITE_FULL, snapshot.tradeClass=INFO_ONLY, demotionTags=[]. Conclusion: the trace doesn't go silent — it goes explicit-INFO_ONLY; the query grabbed the wrong population. The DEMOTED-16 rows are elsewhere. Proposed corrected query (decision='EMITTED' AND reason_code='DEMOTED', LIMIT 20) with a two-way fork on demotionTags, plus a 7-day aggregate by decision/reason_code/setup_key. Bright lines restated.

CLAUDE

Approved both queries but reordered: the 7-day AGGREGATE runs FIRST — it distinguishes three diseases that look identical (setups emit but get demoted; setups die pre-emission; setups are never generated at all). Tightened the fork to three-way: tags populated WITH threshold+actual values → surface-only; labels-only → enrich payload in existing snapshot jsonb; empty → instrument from scratch. Added regime to the aggregate grouping to detect a structural deadlock (trend setups only generated in regimes that auto-veto them — a P1.2 finding). Preserve any "demotionTags=[] on every DEMOTED row" artifact verbatim as the smoking gun.

DEVENDRA — Raw psql output document + "Read again and answer"

Attached the actual -x psql output of the 6 rows (NIFTY/BANKNIFTY/SENSEX, 2026-07-15, confidence 45, RANGING regime, spot/entry/stop/targets populated, option fields NULL, drivers: Spot>EMA21, EMA9>21, RSI>50, VWAP omitted for zero-volume index candles; vix values –4.51 / –4.65 / –4.80).

CLAUDE

Read the raw rows line-by-line and surfaced two things the summary missed. (1) vix = –4.51/–4.65/–4.80 — India VIX cannot be negative (strictly positive, typically 10–25); almost certainly day-change-% being written into a level-typed field, on rows stamped LIVE_KITE_FULL — exactly the defect class the data-honesty layer exists to prevent. Logged as a separate defect (blocks downstream VIX-gated logic), diagnose-only. (2) The evidence came from psql -h localhost -U nse -d nsescanner — demanded proof this is PROD: SELECT current_database(), COUNT(*), MAX(captured_at), and an explicit statement; if dev, rerun everything against prod. Plan otherwise unchanged: aggregate first with regime, then DEMOTED-20, three-way fork, no code, no tuning.

DEVENDRA — Uploaded FNO_SWING_MASTER_FIX_PROMPT_2026-07-15.txt + coder's proposed plan and 4 questions

Coder proposed: run 4 read-only queries (DB verification; 24h aggregate by regime; DEMOTED-20 detail; paper_trade_fo cross-check confirming 0 opens); asked (1) proceed?, (2) re-upload the doc?, (3) VIX bug log-only or fix inline?, (4) anything missing?

CLAUDE

Read the full 1,778-line doc first. Key discovery: the doc's §16 canonical decision taxonomy (EXECUTABLE/WATCH/DEMOTED/REJECTED/SHADOW_FAIL/DATA_BLOCKED) does not contain 'EMITTED' — so the coder's WHERE clause guesses semantics the evidence contradicts. Rulings: 1(a) run all four with corrections — aggregate window 7 days not 24h, run it FIRST, write the demoted-detail query against enum values the aggregate actually reveals; add SELECT DISTINCT decision, reason_code and map every observed value to §16 canonical or mark UNMAPPED (taxonomy drift is now a first-class finding). 2: neither re-upload nor full paste — diagnostics don't need P0.1/P0.2; paste ONE excerpt now, §16's gate-storage mandate ("every gate must store pass/fail/demote, reason code, values tested, threshold/config version, timestamp") — that IS the Step 1 design spec. 3(a) log-only, tagged as blocking P1.3 volatility-aware targets and P2 IV-bucket reports. 4: taxonomy-drift finding + the standing localhost/nsescanner prod-identity check.

DEVENDRA — "what about F&O fixing?"

CLAUDE

Nothing dropped — sequenced behind diagnostics on purpose. In flight: P0.4 queries (promoted over P0.1+P0.2 because the money-path might be dead). Queued: P0.1+P0.2 as one PR the moment Step 0's report lands; then P0.3 readiness gate; then P1 engine repair SHAPED by the diagnosis (if non-BASELINE setups never emit → fix starts at P1.2 setup detection; if they die at a gate → fix starts there with threshold-vs-actual evidence; P1.3 execution fixes only matter once signals reach execution). Discovered defects (negative VIX, taxonomy drift) logged, not fixed inline. First fix PR is one report away. Offered to pre-draft the P0.1+P0.2 directive.

PART 2 — FUNNEL FORENSICS, TAXONOMY RULINGS, INSTRUMENTATION STAGES

DEVENDRA — Step 0 diagnostics complete ("behave like a pro professional trader")

Findings: (1) TWO drop-offs — 50 EMITTED reasoning rows in 7 days but only 6 rows in option_signal_history (~44 vanish before persist); of the 6 persisted, 5 STALE_TRIGGER and 1 triggered-then-expired-without-exit; paper_trade_fo empty LIFETIME (0 rows). (2) Doc §16 gate-storage unmet — values_tested NULL across all 159 rows (score, ivr, ivp, spread%, maxloss% all null). (3) Taxonomy drift confirmed — PRE_EMISSION_REJECTED / EMITTED / SKIPPED vocabularies; reason_code=DEMOTED is a category violation. (4) VIX bug wider: negative on RANGING (–4.70 avg) AND implausibly low on EXPIRY_DAY (~3.2 vs real 13–16). (5) option_signal_history schema thin: no paper_trade_id / execution_status / execution_blocked_reason. Asked: (a) draft Step 1 schema spec first, (b) one more read-only pass tracing the 44 missing via fingerprint anti-join, (c) both b→a, (d) other.

CLAUDE (trader lens)

Answer (c) b-then-a — but the anti-join as drafted would lie: the 50 must first be SEGMENTED by setup_key and tradeClass, because BASELINE INFO_ONLY broadcasts are DESIGNED not to persist; the true leak is only tradeable-class emissions that failed to persist (possibly 4, not 44). STALE_TRIGGER 5-of-6 flagged as the most tradeable finding: entries demand ~0.1–0.14% breakout past spot in a self-labeled RANGING regime — stale by design; compute spot-to-entry distance and emission-to-stale time for all 6, tag to P1.2 evidence. The triggered-no-exit row = P1.3 lifecycle-hole category proof. Schema spec must promote trade_class to a real column (the /audit funnel must never sum INFO_ONLY with tradeable). taxonomy_mapping presented as a proposal table with ambiguous values marked UNMAPPED — mapping is an owner decision. Zero-trades-lifetime framed bluntly: the platform has never completed one signal→trade→exit→P&L loop; the 6–12-month expectancy clock hasn't started.

DEVENDRA — B2–B8 query results (raw psql doc) + sign-off questions

B2/B4 segmentation: 50 EMITTED = 16 BASELINE/INFO_ONLY/DEMOTED + 16 BASELINE/INFO_ONLY/EMITTED + 18 TREND_CONTINUATION rows with NULL tradeClass/reason_code/fingerprint. B3: tradeable-class emissions with fingerprints = 0 rows. B4 also showed PRE_EMISSION_REJECTED breakdown: CONDITIONS_NOT_MET 47, POST_CLAMP_RR 22, LATE_SESSION_ENTRY 12, OTHER 11, NO_LIVE_KITE_INTRADAY 9, VWAP_RECLAIM_LATE 3; SKIPPED/INFO_ONLY_NOT_TRADEABLE 5. B5: no fingerprint column in history. B6: 6 history rows all BASELINE, EXPIRED, 5×STALE_TRIGGER + 1×EXPIRED_TRIGGERED. B7 geometry: bullish entries +0.115–0.143% above spot, bearish –0.095–0.144% below; holds 46–141 min; regimes RANGING/EXPIRY_DAY; avg vix at emit –4.7 / ~3.2. B8: the EXPIRED_TRIGGERED row — 29 min to trigger, 112 min held, exit_price 57448.50, MFE 80.05 / MAE 167.20. Questions: 1 schema sign-off scope (a full / b hold new tables / c changes / d hold); 2 taxonomy mapping decisions; 3 confirm finding tags; 4 next actions.

CLAUDE

Declared the case cracked: the entire real leak is the 18 TREND_CONTINUATION rows — the ONLY tradeable-class lane emitting, 100% lost with partial writes. Everything else explained (104 pre-emission rejections with POST_CLAMP_RR the biggest killer; 32 honest INFO_ONLY broadcasts; 6 persisted; 0 opens). Spotted that B8's exit_price 57448.50 is a SPOT value in an option-premium field (option entry was ~1054) — direct doc-§2 violation, data-honesty class. Rulings: 1(b) — ALTER the two existing tables only; hold both new tables; FIRST run one read-only query pulling all 18 TREND rows (full snapshot, captured_at, clustering, second-writer plausibility). Taxonomy: NO retro-mapping — canonical values for NEW writes only, historical stays UNMAPPED; reason-based override NO_LIVE_KITE_INTRADAY→DATA_BLOCKED wherever it appears (including 9 under PRE_EMISSION_REJECTED); PRE_EMISSION_REJECTED→REJECTED otherwise; SKIPPED per proposal; DEMOTED/EMITTED-as-reason = writer bugs, UNMAPPED + ticket; OTHER banned in new writes. Findings: all confirmed; lifecycle hole upgraded to data-honesty class and becomes P1.3's acceptance replay case; the 18-row finding retitled P0.4-CRITICAL. Next: 18-row query → ALTER(1b) → \d+ paste → stop.

DEVENDRA — Step 1 ALTER complete

9 additive columns on fno_signal_reasoning (gate_name, verdict, stage, values_tested_json, threshold_json, config_version, trade_class, canonical_decision, canonical_reason); 5 on option_signal_history (signal_fingerprint, paper_trade_id, execution_status, execution_blocked_reason, writer_version). Both new tables held. Zero data writes; all columns nullable; indexes preserved. Findings log formalized incl. C1: the 18 TREND_CONTINUATION rows are a placeholder stub — single index/direction, hardcoded signal_date=2026-05-17, null snapshot/fingerprint, tier=STANDARD unique. "The only tradeable lane has NO production writer."

CLAUDE

Reframed the whole case: the platform has NEVER generated a single production tradeable signal — there was never a demoter to instrument; the top of the funnel was never built. Three consequences: (1) new most-important unknown — what are the 104 PRE_EMISSION_REJECTED rows evaluating? If they carry real setup_keys, detection+gates EXIST and P1.2 = wiring (days); if all BASELINE/null, P1.2 = building detection (weeks). One query settles it. (2) The stub must be quarantined, not left polluting prod (~18 garbage rows/week hardcoded to a stale date). (3) P1.2 changes character — includes BUILDING the trend writer; estimates revisit after the setup_key answer; explicit bright line: do NOT "quickly wire" the trend lane during P0.4. Step 2 directive: setup_key×reason_code breakdown + samples → stub quarantine proposal (call-sites, disable vs label, no implementation until picked) → writer-extension spec on paper (how existing writers populate the 14 columns; canonical on new writes only; OTHER banned) → my review before any writer code. No backfill ever without separate approval.

DEVENDRA — "THE STUB IS A TEST-FILE LEAK INTO PROD" (full document)

fnoObservability.test.ts lines 28–36: BASE_PAYLOAD with decision EMITTED, signalDate 2026-05-17, NIFTY, TREND_CONTINUATION, BULLISH, STANDARD, confidence 72; lines 65–94: when DATABASE_URL set, the test WRITES to prod via logFnoReasoning. Every CI/dev/pnpm-test run with env set = one prod row; ~9 runs/day × 2 days = 18. Quarantine proposal: Option A force-mock (delete DATABASE_URL in test, try/finally restore — recommended) vs Option B test-marker + afterAll cleanup. Retroactive cleanup options: delete 18 by signature (recommended) / label writer_version=STUB_QUARANTINED / leave+filter. Full writer-extension spec on paper: column-by-column contracts for all 14 columns (gate_name enum, verdict PASS/FAIL/DEMOTE/SKIP/N-A, stage enum, values_tested_json/threshold_json, config_version bootstrap fno-config-legacy-v0, trade_class TRADEABLE/WATCHLIST/INFO_ONLY/DIAG, canonical fields never NULL, OTHER banned fail-loud), history columns (fingerprint from computeSignalFingerprint, paper_trade_id only by paper-writer at open keyed on fingerprint, execution_status lifecycle, execution_blocked_reason enum, writer_version bump to paper-writer-v1.3.0). Asked: quarantine method, cleanup, spec approval (full vs staged), P1.2 re-sizing (detection RUNS correctly per D1/D2 — days not weeks), order of operations.

CLAUDE

Elevated the real defect: test processes holding PROD write credentials at all — dev/CI DATABASE_URL points at prod; fixing one file doesn't fix the class. Rulings: 1(a) plus a writer-boundary guard — logFnoReasoning throws loudly under VITEST/NODE_ENV=test unless ALLOW_TEST_DB_WRITES=1; prefer db-module mock over env mutation under --pool=threads (flake risk); new ENV-ISOLATION finding: read-only inventory of every environment carrying the prod URL; remediation its own ticket. 2(a) export-first: archive all 18 rows, DELETE ... RETURNING id in a transaction, commit only if count=18, paste both. 3(b) staged (types+mapping helper+unit tests → checkpoint → call sites) with amendments: verdict N/A → NOT_EVALUATED; execution_blocked_reason extensions need owner sign-off; add trade_class='DIAG' as the structural home for future test writes, excluded by /audit. 4 confirmed: P1.2 = days for wiring+instrumentation; §15 validation and tuning stay outside the estimate. 5(a) quarantine → cleanup → staged implementation. Plus: rotate the nse DB password (it's been echoed in plaintext repeatedly) and stop echoing it.

DEVENDRA — Phase A+B complete

Quarantine landed (vi.mock, no env mutation); guard armed; suite 3465/3465 (Δ+1 guard test). Cleanup: 18 rows deleted count-guarded (GUARD_OK exactly 18; ids listed), forensic archive JSON on disk, rows 159→141. ENV-ISOLATION inventory: prod URL in backend/.env, run_postgres.sh, PRD.md, test_credentials.md; NO CI configs; no separate test DB; 17 test files gate on !includes("dummy"). Password hygiene: .pgpass in place; rotation deferred to owner ticket. Spec amended with the 3 amendments. Asked: start Stage 1 now or hold?

CLAUDE

(a) proceed now. Three Stage-1 requirements: (1) mapping-helper tests must encode every precedence ruling as NAMED cases (NO_LIVE_KITE_INTRADAY→DATA_BLOCKED overriding PRE_EMISSION_REJECTED; SKIPPED fork; legacy DEMOTED/EMITTED reasons → asserted refusal/UNMAPPED; OTHER → throws; null/empty → defined+asserted); (2) all new enums as closed TS unions, no string widening in writer-facing signatures; (3) checkpoint evidence = literal full-suite lines + explicit zero-writer-files statement + file list. Owner-side items noted: password rotation post-slice; post-rotation, memory docs should carry the future test-DB URL, prod granted explicitly.

DEVENDRA — Stage 1 checkpoint complete

3502/3502 (Δ+37 = exactly the new taxonomy tests), typecheck 0, fnoCanonicalTaxonomy.ts + test created, zero writer/call-site files touched (grep-verified). Asked: (a) approve → Stage 2 (payload type +9 fields, extend logFnoReasoning, 4 call sites in paperTradingFO.ts + batch dispatch, persistOptionSignalRow history writes, writer version bump, feature flag REASONING_WRITER_V2_ENABLED default OFF), (b) hold, (c) changes, (d) other.

CLAUDE

(a) approved with four conditions: (1) FIRST deliverable = full grep inventory of every logFnoReasoning caller mapped to extended / covered-by-batch / explicitly-skipped-with-reason — the pre-emission writers carrying the 104-row volume must be covered; (2) flag OFF = byte-identical behavior (new columns NULL), fail-loud non-null asserts only when ON, tests cover both states; (3) checkpoint = full-suite + typecheck + both-flag tests + file list → earns flag-OFF deploy only; (4) flag-flip is a separate step whose acceptance is a read-only query on the next live session (all mandatory columns non-null, zero OTHER, values_tested_json real numbers). Bright line: nothing wires or activates the trend-lane paper path.

DEVENDRA — Call-site inventory + 3 clarifications

5/5 reasoning writers + 1 history INSERT + 6 UPDATE sites; Sites D (buildEmittedRow — BASELINE broadcasts) and E (pre-emission) carry 100% of volume. All 5 extended, nothing skipped, trend lane not wired. Questions: (1) batch dedupe keyed on legacy (decision,reason_code) with canonical shadow (a) vs widened (b); (2) Site D writes legacy reason_code='DEMOTED' when demotionTags>0 — keep legacy + canonical override with UNMAPPED (a) vs delete legacy line (b) vs let helper throw noisily (c); (3) F/G/H UPDATE sites lack paper-awareness — split responsibility (a: they set execution_status, paper-writer backfills paper_trade_id) vs hold paper_trade_id (b).

CLAUDE

1(a) confirmed — canonical fields are deterministic derivations of the legacy key; widening adds zero information. 2(a) AMENDED: don't discard information in hand — canonical_reason = the primary demotionTag mapped to canonical taxonomy where clean; where unmappable use a NEW specific enum LEGACY_DEMOTION_UNMAPPED (not generic UNMAPPED) linked to the writer-fix ticket; the tag→canonical mapping table comes for sign-off at the Stage 2 checkpoint. (b) rejected — breaks legacy consumers; (c) rejected — expected-input warnings train people to ignore warnings. 3(a) confirmed with an honesty guard: a writer-permission matrix — each site may only write execution_status values it can verify from its own data; no sweep writes TRIGGERED_OPEN without paper-open awareness; sites that can't determine truth leave NULL; propose intermediate enum values for sign-off if needed.

DEVENDRA — Stage 2 checkpoint complete

3548/3548 (Δ+46: 31 taxonomy + 15 writer-flag), typecheck 0, clean boot. Flag defaults OFF with 4 byte-identical assertion tests. 9 files touched. One schema hotfix disclosed: paper_trade_id was bigint but paper_trade_fo.id is UUID string — corrected to varchar(64) via ALTER COLUMN TYPE on the zero-row column. Sign-off items announced (13-tag mapping table; 4-writer permission matrix). Asked flag-flip timing: (a) flag-OFF deploy, flip later; (b) flip now (market open); (c) changes; (d) hold.

CLAUDE

Three points before the flip: (1) the hotfix was the RIGHT fix through the WRONG door — ALTER COLUMN TYPE is not additive; standing rule restated: any schema statement not literally ADD COLUMN IF NOT EXISTS gets checkpointed first, zero-row columns included; noted, not undone. (2) Counts aren't sign-off — paste both tables in full. (3) Timing: neither (a)-as-stated nor (b) — tomorrow (Fri) is a trading day, so no Monday wait; but no behavior-change deploys during market hours EVER (discipline as asset): confirm flag-OFF restart now, flip after 15:30 IST today, acceptance runs on Friday's session. Sites A/B/C verification logged as an OPEN item that closes with P1.2's first real paper trade — not claimed now.

DEVENDRA — "." (empty ping)

CLAUDE

Status summary: ball with the coder — awaiting the two tables inline, flag-OFF restart confirmation, then the post-15:30 flip with Friday acceptance. Owner list: password rotation + memory-docs URL decision post-slice. Next decision point: reviewing the mapping table as a trader (e.g., HTF-conflict tag must not be lumped with liquidity rejections). Queue after Friday: 2–3 session soak → /audit panel → P0.1+P0.2 first fix PR.

PART 3 — DEPLOY, DRIFT-P0, CODE REVIEW, BRIEFING FEATURE

DEVENDRA — Acknowledgements + full sign-off tables + deploy plan (document)

(1) Schema-rule breach acknowledged verbatim; future path discover→propose→wait→execute. (2a) Demotion-tag→canonical mapping, order-sensitive, 13 rows: HTF_CONFLICT & HTF1H_CONFLICT→HTF_BIAS_CONFLICT; OI_ATM_CONFLICT; RS_CONFLICT; COUNTER_TREND→COUNTER_TREND_BIAS; EXPIRY_DAY→EXPIRY_DAY_MODE; OPENING/CLOSING_NOISE→_WINDOW; VOL_CLAMPED_STOP; RECOVERY_MODE_VETO; CHASE_RISK_VETO; RR_LOW→RR_INSUFFICIENT_POST_CLAMP (shared bucket with pre-emission POST_CLAMP_RR); LOW_WINRATE→LOW_WINRATE_HISTORY; escape → LEGACY_DEMOTION_UNMAPPED. Precedence: structural/data > directional > regime > session windows > portfolio vetoes > geometry > statistical. (2b) Writer-permission matrix: PAPER_WRITER all six statuses; LIFECYCLE_SWEEP NOT_TRIGGERED/AWAITING/CLOSED only; KITE_TICK_SWEEP NOT_TRIGGERED/AWAITING; ORCHESTRATOR_HOOK NOT_TRIGGERED only; BLOCKED/ERROR paper-writer-exclusive; fail-closed NULL. Asked: rule confirmation; table sign-offs; flip timing (15:44 IST, post-close); Sites A/B/C open-item wording.

CLAUDE

1(a) confirmed. 2(b): mapping table approved AS-IS (with the understanding the full demotionTags array keeps writing to snapshot — winner for aggregation, array for forensics); matrix has ONE honesty hole that is exactly the B8 class and WILL recur Friday: LIFECYCLE_SWEEP permitted TRIGGERED_CLOSED, but a spot-based sweep cannot truthfully assert a close of an open it never observed — and with zero paper trades, EVERY triggered+expired signal hits this. Two-part fix: state-transition guard (sweep may write TRIGGERED_CLOSED only when current status == TRIGGERED_OPEN) + one pre-approved new enum TRIGGERED_EXPIRED_UNEXECUTED (sweep-writable terminal: trigger fired, never executed, lifecycle ended). 3(a) green light sequenced AFTER the matrix fix: implement guard+enum, tests, paste, then flip tonight; Friday acceptance adds the assertion: every triggered-and-expired signal shows TRIGGERED_EXPIRED_UNEXECUTED — never TRIGGERED_CLOSED, never NULL-after-trigger. 4(a) approved.

DEVENDRA — Flag ON deployed (16:17 IST, 47 min post-close)

3556/3556, typecheck 0, apiserver clean; B8 class foreclosed via TRIGGERED_EXPIRED_UNEXECUTED + canLifecycleSweepCloseFrom; 18 stub rows deleted; writer test-env guard armed; Friday acceptance query drafted (7 sections incl. B8 strict check). Next items listed: run Friday acceptance; build audit panel after soak; fix trigger geometry; wire real trend writer.

CLAUDE

Deploy accepted. Two sequence corrections: (1) P0.1+P0.2 slots INTO the soak window — Friday acceptance closes Step 2 → Monday P0.1+P0.2 starts → /audit panel after soak; don't let the panel or P1.2 queue-jump. (2) "Fix Trigger Geometry" is mis-framed — it's the OWNER's strategy decision; the coder's P1.2 deliverable is an options memo with evidence, never a unilateral threshold pick. Added a Friday preamble sanity query (row count + time range) and the thin-sample rule: extend through Monday if volume is thin.

DEVENDRA — Sequence corrections applied (PRD + query file edited) + Publish / Run Code Review buttons

CLAUDE

Hold one beat before Publish — the schema landmine lives exactly there. Gate 1: paste the drizzle-kit push diff vs nsescanner — must be ZERO pending; all 14 columns declared in TS with matching types (paper_trade_id varchar(64)); if not empty, stop and reconcile (post-merge push must be a no-op by construction — the strategyControl.ts lesson). Gate 2: state whether Publish redeploys the running apiserver; if yes, show REASONING_WRITER_V2_ENABLED=1 in the published deployment's env so Friday doesn't silently run flag-OFF. Code Review: yes unconditionally — read-only, free, and this slice touched the writer boundary of a financial audit trail.

DEVENDRA — Publish HELD: both gates failed

Drizzle diff: 14 columns in parity BUT pre-existing drift would DROP 5 tables (daily_report_runs, system_alert_dedup, system_alert_state, notification_delivery_log, reconciliation_report) + 3 constraint changes — runtime CREATE TABLE IF NOT EXISTS tables undeclared. Deploy env: Emergent Publish reads deployment-scoped config, NOT backend/.env — flag would not carry. Notes: Friday's acceptance runs against the preview pod (what the live session hits) — Publish NOT required. Options: A defer Publish (recommended); B fix drift on paper tonight then publish; D declare only the 5 missing tables tonight.

CLAUDE

Option A — defer. B fails on time-pressure (8 declarations at 5pm for a deadline that doesn't exist is how a TS type subtly mismatches a live column — the bigint/varchar bug proved the class); D fails its own premise (5-of-8 still isn't a clean push, so Publish stays blocked anyway — all of B's risk, none of its payoff). Registered: DRIFT-P0 — until reconciliation lands, NO merges triggering post-merge.sh and NO manual drizzle-kit push, no exceptions without explicit approval (current diff would DROP 5 prod tables). Reconciliation slice next week: origin-story inventory of all 8 items → TS declarations matching live DB byte-for-byte → zero-pending diff → deployment-env audit incl. the flag → Publish. Run Code Review now against preview state (real defects checkpoint before fix; style log-only).

DEVENDRA — Code Review results (full document) + placement question for defects #1/#2

Findings: #1 HIGH — race: post-open TRIGGERED_OPEN history stamp is fire-and-forget outside the txn; a concurrent sweep can write TRIGGERED_EXPIRED_UNEXECUTED then be overwritten by the late stamp → fabricated open on a closed position (B8 re-emergence). #2 MED — paper close path never stamps TRIGGERED_CLOSED (no post-close stamp; closed trades stay TRIGGERED_OPEN forever). #3 MED — DB read type-asserts executionStatus to the union without runtime validation (accidentally fail-closed, silent). #4 LOW — fingerprint 64-bit truncation: comment math wrong (~0.66% collision at 365k rows, not 1e-10). #5 LOW — no endpoint exposes flag state; getReasoningLoggerHealth built but mounted nowhere. Style: redundant cast idiom; magic 16 truncation constant. Observations: fresh env evaluation correct; B8 NOT reintroduced (sweep exitPrice is spot-based BY DESIGN per the spot-based lifecycle FSM; paper close writes real premium); Site E OTHER-swallow intentional; flag-OFF byte-identical enforced; transition guard applied but never fires in practice due to #2. Question: fold #1+#2 into (1) a new PAPER_WRITER discipline mini-slice before P1.2, (2) the drift slice, (3) P1.2's opening moves, (4) other.

CLAUDE

Option 1 — dedicated PAPER_WRITER-DISCIPLINE mini-slice after drift reconciliation, landing BEFORE P1.2 (hard prerequisite: no real paper trade may ever open against the current race). Not the drift slice (its safety argument is "zero behavior change" — don't contaminate); not P1.2 (already the biggest slice; fixes deserve their own checkpoint). Reassurance: neither defect threatens Friday — paper path can't fire without the trend lane. Fix shape mandated: root cause is app-level read-then-write, so move legal-transition enforcement INTO the SQL — one transitionExecutionStatus(fingerprint, from[], to, writerId) compare-and-set helper (UPDATE ... WHERE status matches expected priors; stale writers match zero rows), permission matrix + transition table enforced inside it, all four sites converted; #2 becomes just another transition call; #3 rides along (helper validates reads against the closed union and LOGS anomalies). #4: widen to varchar(64) full-hash for new writes in the same slice (pre-approved) + fix the wrong comment. #5: folds into Step 3's /audit panel (mount health owner-only + flag state). Queue: Fri acceptance → P0.1+P0.2 (soak) → drift → PAPER_WRITER-DISCIPLINE → /audit → P1.2.

DEVENDRA — Defect #3 placement options (drift slice / discipline slice / defer)

CLAUDE

Option 2, per the prior directive — #3 rides inside the transitionExecutionStatus helper (three lines in code being written anyway); putting it in the drift slice would break that slice's zero-behavior-change safety argument. Precision kept: current cast is ACCIDENTALLY fail-closed — defect is silence + fragility, not unsafety today; fix = validate + warn-log (fingerprint, raw value, writerId) + refuse the transition — never throw mid-sweep. Option 3 (defer) rejected: "accidentally safe" is not a property to carry into P1.2's real traffic.

DEVENDRA — Defects #4/#5 + 2 style items (log-only / fix now / other)

CLAUDE

Both already dispatched; restated precisely. #4 split: wrong comment fixed opportunistically (zero-risk doc edit); the widening is PRE-APPROVED into the discipline slice. #5 scheduled into Step 3's panel (a standalone mount tonight = behavior change on the pod the night before acceptance, for zero Friday benefit). Style items genuinely log-only — absorbed by the same fixes (#3's helper deletes the cast line; the constant extraction lands with #4). Rejected "fix now" as a batch: small-fixes-tonight on the frozen acceptance pod is the adjacent-action pattern; the freeze is now the operative rule until Friday's results are in.

DEVENDRA — "Anything from the code review you want addressed differently?"

CLAUDE

Yes, two — both cases where the review's CORRECT observations undermined earlier conclusions. (1) B8 re-characterized: the sweep writing spot values is BY DESIGN (spot-based FSM), so the defect isn't "spot leaked into a premium field" but "ONE column carrying TWO unit semantics with zero provenance" — P1.3's fix becomes splitting the column (spot_exit_level vs option_exit_premium) or a unit label, not guarding the sweep. Amended the P1.3 finding text. (2) The OTHER-swallow makes Friday's "zero OTHER" trivially true — the honest metric is the UNMAPPED count; added an acceptance section: UNMAPPED canonical_reasons grouped by gate_name and legacy reason_code (quantifies the writer-fix ticket on live data). SQL-file edit, not a pod change — freeze intact. Minor third: health counters mounted nowhere means a dead writer mid-session goes unseen; approved a read-only ~12:00 IST spot-check Friday.

DEVENDRA — PRD updated; pod frozen summary (17:15 IST) + "Is the coder going in the right direction? Are we addressing the real issue?"

CLAUDE

Yes on direction — strongest stretch yet (self-corrections on evidence, no-code line held, drift caught BEFORE Publish, breach owned without defensiveness; every finding a real landmine). Caveat: two real issues exist and only one is being resolved. Problem 1 (the platform lies to itself): largely RESOLVED — honesty layer, provenance, instrumented funnel, fabrication classes foreclosed. Problem 2 (the platform has never proven it can make money): NOT STARTED — zero trades lifetime; the truth-infrastructure is comfortable work that can't answer the expectancy question. Critical path: drift → discipline slice → P1.2 trend writer + owner geometry decision. Watch items: process must not become the product (queue should get FASTER per slice from here); the 6–12-month out-of-sample clock is the real deadline and starts only at the first honest trade.

DEVENDRA — Same frozen-pod summary re-sent

CLAUDE

Confirmed registered; calendar: Fri — 12:00 canary → close → 9-section acceptance after 15:30 → paste; clean closes Step 2, thin extends to Monday. Mon — P0.1+P0.2 regardless. Then drift → discipline → panel → P1.2 with the geometry memo first. Nothing needed from Devendra before Friday evening. Reading guide for the results: the UNMAPPED-by-gate breakdown and whether values_tested_json shows real numbers on POST_CLAMP_RR rejections — raw material for the eventual tuning decision on the funnel's biggest killer.

DEVENDRA — Uploaded Daily_Market_Analysis_Prompts (2).md + "planning next step: see attached prompt"

(Pre-market and post-market professional analysis prompt templates for NIFTY/BANKNIFTY/SENSEX — full text in Appendix B.)

CLAUDE

Right kind of next step (improves the trader, not the coder's queue) — but split into two tracks. Track 1: use the prompts MANUALLY with Claude now, zero cost — especially §11 trade-journal tie-in (targets the emotional-override pattern) and §6 level validation; manual use also pressure-tests the template before code exists. Track 2: the wired-in auto-briefing is real platform work slotted AFTER P1.2 — because (a) it would delay the first real trade, (b) its best sections need data the platform provably lacks (participant OI → bhavcopy blocked from prod IP; GIFT/DXY/US10Y not in Kite and the US10Y parser had the 0.45-vs-4.5% defect; internal VIX column corrupted), (c) the natural home is EXTENDING the existing Telegram reports, section-gated on each data source being fixed, with provenance labels. Registered: manual routine now; auto-briefing post-P1.2 with named dependency gates. Offered to inaugurate the post-market prompt after Friday's close.

DEVENDRA — "I need a prompt for the coder to implement it + a Telegram template for pre/post market wrap. Do not miss anything, professional, full-fledged, feature-rich. This is for my Pre Post tab."

CLAUDE

Delivered PRE_POST_BRIEFING_MASTER_PROMPT.md (full text in Appendix C) with three baked-in guardrails: DRIFT-P0 forbids merging (build on a branch); SECTION-GATING as the core design (every section carries provenance state ACTIVE/PARTIAL/GATED, ships honestly as UNAVAILABLE until its source is trusted, unlocks individually — no section may fabricate); Phase 0 is a read-only evidence-first Data Availability Matrix before any code. Covers all 10 pre-market + 11 post-market sections, two full Telegram templates (HTML mode, 4096-char part-splitting, provenance chips ■■■■✍■), Pre/Post tab spec (dark-terminal, journal input, owner controls), 2–3 new tables proposal-first with same-day Drizzle declarations, IST scheduler with loud-partial failure behavior, 6-phase checkpointed delivery, explicit do-not list, 3-day definition of done. VIX sourced ONLY from a direct live Kite quote with a 5–60 sanity band — never the corrupted column.

PART 4 — BRIEFING PHASE 0, THURSDAY CLOSE, FRIDAY LIVE-MARKET P0

DEVENDRA — Coder absorbed spec; 4 scoping questions + DRIFT-P0 carve-out flag

(1) Phase 0 timing vs Friday freeze — recommended executing Thursday evening; (2) branch name; (3) matrix probe ordering positive-first vs gated-first; (4) PRE-8 News interim: MANUAL owner-notes day one vs fully gated. Sanity checks: Phase 0 is read-only, freeze-compatible; Phase 1's DDL will need drift reconciliation landed OR a carve-out for born-declared tables.

CLAUDE

1(a) tonight — with a hard boundary (row F/G = plain fetch from current env ONLY; the failure IS the deliverable; no proxies/header games) and a constraint (row K's rate sweep never during market hours — it shares the Kite session with signal generation on acceptance day). 2(a) feature/pre-post-briefing. 3(a) positive-first — the ACTIVE-set confirmation is what Checkpoint 0 signs. 4(a) MANUAL notes day one — same input primitive as the POST-11 journal (build once, mount twice: morning intent + evening review = the pre-commitment loop against emotional overrides); hard rule: MANUAL text renders verbatim under ✍■ only, never feeds any computed line. Carve-out RULED: approved in principle — Phase 1 may apply the 3 briefing DDLs + same-day Drizzle on the branch, conditional on a branch-diff proving zero additional pending items beyond the known 8; the MERGE stays blocked under DRIFT-P0, no exceptions.

DEVENDRA — Plan doc committed; anchor files identified (dailyReports.ts, marketData/router.ts, optionChainSnapshotIngestor.ts, alerting.ts, telegramBotCommands.ts); next-actions list said "Run Phase 0 Matrix Friday morning"

CLAUDE

Caught the slip: Phase 0 was ruled TONIGHT (Thursday), not Friday morning — stacking 12 evidence rows against a hard 09:00 deadline on a four-deliverable day is how rows get "confirmed" from memory. Tonight, or explicitly Friday post-close with row K — never "Friday morning." Second: optionChainSnapshotIngestor.ts already existing is a scope discovery — added matrix row M: document what it captures, its schedule, backing table, and Drizzle/drift status; if it covers PRE-6/POST-5, Phase 1's schema shrinks.

DEVENDRA — Phase 0 first evidence pass; TWO scope-shift surprises

Row M: ingestor ALREADY LIVE, both backing tables Drizzle-declared (not in drift set) — Phase 1 shrinks to 2 new tables + reuse. Row G: cash bhavcopy HTTP 200 (370KB) and FII/DII API HTTP 200 with today's real numbers (DII +■2,986.41 cr, FII –■4,205.56 cr) BOTH work from prod IP — proposing three sections flip to ACTIVE. Confirmed gated: row F participant OI

(404), row H GIFT absent.

CLAUDE

Good news taken skeptically: row G contradicts a documented standing blocker, and contradictions get investigated. Four candidate explanations (NSE change / cash-vs-F&O endpoint split / header-session differences / off-hours-only access) — the matrix must record which the evidence supports. Classification: PROMOTABLE, not ACTIVE — trust for an external fetch = works from the deployed env, at the scheduled time, across multiple days; promotion requires a 3-day fetch-reliability record running passively during Phases 1–2, complete before Phase 3 wires Telegram. Row F's 404 may be publish-timing (participant files land late evening) — retest after 20:00 before calling it gated. Capture tonight's FII/DII endpoint + response timestamp + update cadence as evidence for the 18:45 POST_FULL schedule. Row M accepted; document the ingestor's schedule vs the 09:20/15:25 snapshot needs.

DEVENDRA — "Welcome back" state read + lane choice (a: continue matrix; b: run Friday acceptance if past 15:30 Fri; c: drift planning; d: PAPER_WRITER design memo; e: other)

CLAUDE

Lane (a) — and calendar correction: it is THURSDAY evening; the acceptance query is ~24h away. Tonight: rows C/A/B/D/E/I/J/L read-only, row C (direct NSE:INDIA VIX quote + sanity band) first; row G why-investigation; row F retest after 20:00. Hold (c) and (d) — one-slice discipline; the drift inventory starts after Monday's kickoff; the discipline memo's core primitive is already specified in the PRD. Confirmed the queue is a STRICT SEQUENCE, not a priority pool: acceptance → P0.1+P0.2 → drift → PAPER_WRITER-DISCIPLINE → /audit → P1.2.

DEVENDRA — Row F retest logistics (stay online 20:15 / Claude runs it / roll to Friday / other)

CLAUDE

Option 3 — roll into Friday's post-close docket with row K and the row-G re-run: one batch, one methodology, realistic fetch window; Friday's file is BETTER evidence (fresh trading day, scheduler-realistic window). Rejected keeping a session alive 90 minutes for one non-blocking data point, and rejected splitting probe provenance across two actors. Before closing: mark rows F/G in the matrix as OPEN — RETEST FRI POST-CLOSE with tonight's partial evidence attached.

DEVENDRA — VIX-corruption escalation (48h = 30 NULLs + 6 negatives): P1-after-drift / escalate-P0-now / never-fix-just-avoid / other

CLAUDE

Option 4 (hybrid): DIAGNOSE now read-only, FIX stays in sequence. Escalation fails the "what changed" test — re-observing a known, contained, non-propagating defect isn't new exposure (nothing live reads the column; briefing is spec-barred from it). Avoidance is rot — P1.3 and P2 are scheduled consumers. Diagnostic added to the docket: grep every write path into fno_signal_reasoning.vix; identify the actual value passed (test the day-change-% hypothesis against real India VIX for Jul 14–15); explain the 30-NULL vs 6-negative split. Fix slots as a rider on PAPER_WRITER-DISCIPLINE (post-drift, pre-P1.2 — P1.3 needs trusted VIX before the first real trade). History stays dirty with a documented cutover date, consistent with no-retro-mapping. Stated the general rule: re-observing a known defect doesn't re-prioritize it — new exposure does.

DEVENDRA — Row J News flip question: 20 Indian RSS feeds already wired via newsRss.ts — ACTIVE+MANUAL in v1, or hold pending source audit?

CLAUDE

Hold — and not primarily for the source-list reason: "20 feeds fetch successfully" answers the wrong question. PRE-8/POST-8 are editorial artifacts; raw RSS is an unranked headline stream, and SELECTION is where a briefing goes quietly dishonest (whichever 5 headlines the code picks get the platform's credibility stamp). Classification: PROMOTABLE-infrastructure / GATED-as-section. Promotion path: (1) owner audits the 20-feed list; (2) curation design decision — default preference a dumb-but-honest v1.x (whitelisted OFFICIAL feeds only, latest-N timestamped, labeled "unranked headlines, not analysis"); (3) split the EVENTS CALENDAR out of row J as its own line — scheduled-events data wants a structured source or owner table, not RSS parsing, and it's the higher-value half. v1 ships MANUAL ✍■ only.

DEVENDRA — Thursday docket COMPLETE summary + agent's 4 suggested next items + Publish + Code Review buttons

Matrix finalized (10/13 rows evidenced; F, G-rerun, K rolled to Friday). VIX fully diagnosed: single writer at optionSignals.ts:3191 passes gateCtx.vix.intradayPct (a change-%) into the level-typed field; VixSnapshot has no .level; fix ≤3 lines + tests, queued as the discipline-slice rider. Row J held; 20-feed list pasted (all aggregators); Row J' Events Calendar split out. Suggested: Friday batch; Row J audit; Events Calendar design memo; VIX fix-preview memo.

CLAUDE

Rulings: NO Publish (DRIFT-P0 unchanged). NO code review — read-only evening, nothing to review (process-as-theater otherwise). Item 1: Friday batch proceeds as scheduled. Item 2 ruled from the pasted finding: the audit FAILS — an all-aggregator list has nothing to whitelist; promotion now requires a PROPOSED official-source list (RBI/NSE/BSE/SEBI/PIB), low priority. Item 3 deferred into Briefing Phase 1 planning post-Checkpoint-0 — no new threads tonight. Item 4 DECLINED — the diagnosis IS the memo; writing the literal diff weeks early is shovel-readiness theater and the adjacent-action pattern in its most innocent costume; the diff gets written inside the discipline slice with its tests. One genuine addition: the rider's acceptance must define behavior for the VIX-unavailable case (the NULLs), not just the unit bug. Session closed; next contact Friday 12:00 canary.

DEVENDRA — Registered; session-close ledger (noting rider covers "6 negatives AND 109 NULLs")

CLAUDE

Clean close; one number flagged for later reconciliation: 30 NULLs (48h window) vs 109 NULLs (full table?) — the rider's acceptance query must state its denominator explicitly. Friday's one-line script confirmed: 12:00 canary → 15:30+ probes → evening acceptance → paste results.

DEVENDRA — FRIDAY ~11:23 IST: STACK DOWN

Coder found Node api-server and Postgres both unreachable; container recycled overnight; supervisor FATAL after 10 retries; apt lock held; postgres error log showed every option_signal_history INSERT failing "value too long for type character varying(32)" on a ~1/min tick since 10:50 IST — signal ledger dead during live market. Asked: (a) restart supervisor postgres now; (b) investigate first; (c) hold; (d) other.

CLAUDE

Option 4 — do NOT restart; the framing misread the incident as one problem when it was two. Problem A: per the coder's own evidence at that moment, postgres appeared UP (pid running checkpoints; diagnostics succeeded) — what was broken was supervisor's VIEW (backoff, apt self-heal loop); restarting the unit mid-session risks an apt reinstall against a live data dir — environment repair is post-15:30. Problem B — the real incident: the varchar(32) overflow; prime suspect checkable in one minute: paper-writer-v1.3.0-reasoning-instrumented is 42 chars vs writer_version varchar(32); first live session with flag ON = first time real column widths were exercised. Directives: confirm the column via \d; EMERGENCY SCHEMA APPROVAL granted in writing for exactly ALTER COLUMN <col> TYPE varchar(64) — catalog-only, no restart, writes resume next tick; logged freeze exception (the freeze protected the measurement; the measurement was dead); same-day Drizzle update; verify log-quiet + post-fix insert count; damage assessment both tables; extend acceptance sample through Monday; post-close docket adds supervisor repair, flag-survival check, and a permanent column-width invariant test. Noted the silver lining: the instrumented writer failed LOUDLY on day one — the failure mode we built.

DEVENDRA — Coder's four options for the overflow (diagnose-only / emergency ALTER + restart apiserver / flag OFF / defer)

CLAUDE

Option 2 minus the restart — the ALTER needs NO restart (per-statement errors; apiserver already self-restarted and doesn't cache widths). (1) rejected: leaves the ledger dead ~4 days of live sessions AFTER the cause is known — chosen data loss. (3) rejected: requires a mid-session restart, destroys the instrumented sample, and un-ships the instrumentation to dodge a width bug. (4) rejected: deferral during active bleed with a 90% root cause and a reversible fix is not caution — with one legitimate stop trigger built in: if \d shows the overflowing value is DYNAMIC (not a fixed literal), stop — the bug would be in what's written, not the width. Sequence: connectivity → \d confirm → ALTER → log quiet → post-fix evidence.

DEVENDRA — New symptoms mid-incident: paper_trade_eq.quantity missing + system_alert_dedup table missing — diagnose alongside or defer?

CLAUDE

Neither — reframe: system_alert_dedup is one of the 5 drift-set runtime tables; it plus a missing column suggests the recycle may have rebuilt/partially re-initialized schema state — which raises whether Wednesday's 14 direct-DDL columns survived, and whether the overflow is "sized wrong" vs "our sizing didn't survive." Mandated a full schema-integrity sweep BEFORE the ALTER: both \d's with widths vs applied values; \dt vs the 41+5 inventory; paper_trade_eq spot-check; drift-set tables present/absent; row counts vs baseline (141 reasoning / 6 history) — count regression = full stop and damage report. Fix then scoped by findings: clean-overflow → single ALTER; missing columns → re-apply the already-approved DDL batch; data regressed → stop. Post-close docket elevated: container-recycle persistence question (what actually survives) and drift reconciliation re-baselined against post-recovery reality.

DEVENDRA — RULING CORRECTION: PG fully DEAD (no process, no binary — postgresql-15 server package uninstalled by the recycle; PGDATA intact 68M; apt lock now clear). Options: (a) recover now then run the authorized sequence; (b) recover then stop after sweep; (c) investigate run_postgres.sh first; (d) escalate platform risk. Also: is today still worth measuring?

CLAUDE

(a) with two amendments: a 2-minute PREFLIGHT from (c) — cat /app/.pgdata/PG_VERSION and confirm the major version matches what apt will install (the one way recovery could damage PGDATA); and a NARROWED auto-proceed from (b) — proceed without pasting ONLY on the clean-overflow branch (all 14 columns present, width-only defect, counts ≥ baseline); any other finding stops for eyes-on. Noted the recycle wiping binaries while keeping PGDATA appears to be DESIGNED behavior (that's what run_postgres.sh's apt-on-boot exists for); the failure was the apt lock starving the self-heal — "make self-heal survive an apt lock + alert on FATAL" becomes a named post-close P0-resilience item; (d) overkill. On measurement: YES, today still counts — 09:15–10:50 is ~95 minutes of the first-ever instrumented live evidence; post-recovery rows extend it; and the incident window ITSELF is acceptance evidence that failures are loud, not silent. Partial sample with an annotated gap = exactly what honest instrumentation produces.

DEVENDRA — "System was live earlier, now not working, I logged in — tell me the Redirect URL" + coder gave URL and asked recover-first or hold

Coder supplied https://kite-analytics-lab.preview.emergentagent.com/api/kite/callback and noted the callback 500s until PG persists tokens; flagged the two pending recovery rulings untouched.

CLAUDE

Option 1 — recover first; ruling already given and standing: preflight → recover → sweep → fix; the Kite login is wasted until PG can persist the session (the just-attempted login almost certainly didn't stick — redo after recovery, then verify kite_session row + KiteReadiness green, which also matters for the afternoon's rows being LIVE_KITE_FULL). Hygiene note: KITE_API_KEY now in the transcript plaintext — add to the credential-rotation ticket. Clock check: ~11:40, close 15:30 — every minute buys measurable session.

DEVENDRA — RECOVERED 11:43:49 IST · bug_testing_agent verdict=fixed

postgresql-15 reinstalled (root cause: run_postgres.sh directory-vs-binary check defect skipped its install branch); ALTER writer_version → varchar(64) executed; Drizzle updated same-day. Evidence: width=64; history 6→9 rows; 3 rows carry the 42-char stamp; 0 overflow errors post-06:13:49 UTC; reasoning writes uninterrupted throughout; Kite KITE_READY (user Devendra, MRV421). Damage window ~54 min (10:50–11:43) of dropped history inserts. Next items: tonight's acceptance with gap annotation; run_postgres.sh glob fix; column-width invariant test; paper_trade_eq.quantity fix; persistence audit.

CLAUDE

Recovery accepted — 20 minutes ruling-to-verified-fix, no restart, zero drift, independent verification. The recovery also answered the morning's scariest question: the "missing tables" were symptoms of PG being DEAD — PGDATA and all direct DDL survived; only binaries die; persistence audit shrinks to a bounded documentation item (one loose end: re-verify system_alert_dedup exists post-recovery). Docket rulings: acceptance tonight with the gap annotated and the 3 post-fix 42-char rows cited as contract round-trip evidence; run_postgres.sh fix approved post-close TODAY (diff + dry-run test plan → go → apply) as highest priority; invariant test approved post-close (must FAIL against the pre-fix declaration to prove it tests reality); paper_trade_eq diagnose-then-propose; persistence note folds into the same pass. Bright-line note for the record: the coder executed exactly the pre-authorized branch and stopped at boundaries — the process had an emergency lane built, and it held under fire.

PART 5 — LEDGER, SIGNAL CARDS, THE MISSED-MOVE DAY, THE COMPLETION MISSION

DEVENDRA — Full open-inventory ledger + BUG-53/54 explanation

Loose end closed live: all 5 drift-set tables verified present post-recovery (the 11:23 to_regclass failures were pure DB-dead symptoms). Post-close docket registered (5 items + Friday probes + acceptance). Full ledger: strict sequence (acceptance → P0.1+P0.2 → drift re-baselined → PAPER_WRITER-DISCIPLINE + VIX rider → /audit → P1.2 → P1.3 → Briefing 1–4); held (Row J official-source condition; Row J'); backlog (ENV-ISOLATION; password rotation; KITE_API_KEY hygiene; 3 flaky globalPresetRoutes tests); dropped (BUG-53/54 — IDs with no recoverable spec from the old numbered Master Fix File; the A–G direct spec now authoritative); on-hold (Phases 7/8/9; Phase B ledger change).

CLAUDE

Cross-check holds. BUG-53/54 drop CONFIRMED with sharpened reasoning: the platform has since been audited FROM EVIDENCE, not memory — the master-fix doc + P0.4 forensics re-derived every materially open defect; lost-spec items are either already captured (that range maps onto P1.2 + writer tickets) or unreproducible; implementing a guessed spec = inventing work. Condition: record the drop with a breadcrumb in the PRD ("spec unrecoverable; superseded by evidence-based audits; diff against ledger if the text ever surfaces"). Endorsed the principle as a standing rule: specs enter the roadmap with FULL TEXT in the repo, never as numeric pointers. One promotion: ENV-ISOLATION rides immediately behind drift reconciliation (same env-file surface; the 17 test files remain a prod-write hazard held by exactly one guard).

DEVENDRA — BUG-53/54 ORIGINAL SPEC SUPPLIED (Trading Desk section) + coder committed + sequence question

BUG-53 Signal cards missing fields — every card must render: tier badge, detector name, direction, index+strike+type, confidence, entry/stop spot AND premium, T1/T2 both units, R:R, lots+notional ■, expected risk/reward ■, all demotion tags, guard tags G1–G4, data_source badge (kite_ws/kite_rest/stale), timestamp with data age. BUG-54 Live signal state — cards show lifecycle live from StateBus: PENDING (current spot · distance to entry · ETA if trend continues), TRIGGERED (entered at · slippage), OPEN (current premium · MTM · distance to SL/T1), CLOSED (exit price/time · realised · reason). Coder committed the full text to PRD per the new rule; sizing notes (Step 2 stamps much of the write side already; unknowns: read-surface exposure, G1–G4 stamping, per-tick data_source, StateBus shape); sequence proposal α (/audit → 53 → 54 → P1.2) vs β (P1.2 → 53 → 54); G1–G4 hard v1 requirement or v1.1?

CLAUDE

Modified β: cards after P1.2 AND after P1.3. Not α — every slice before P1.2 delays the first trade, and BUG-54 is the starkest case (a live-lifecycle overlay over a feed with zero real trades = built against dry runs, re-validated later); BUG-53 pre-P1.2 = polish on an empty feed. P1.3 precedes 54 specifically because CLOSED renders "realised ■" consuming the exact two-unit-ambiguous exit_price column — cards render LAST, on clean columns. Sequence: /audit → P1.2 → P1.3 → BUG-53 → BUG-54 → Briefing. G1–G4: conditional/v1.1 — BUG-53 v1 renders ONLY what the write side already persists (no display slice forces writer work); first deliverable = a field-availability check across all 15 fields; gaps render as "—" with a reason chip. Two spec amendments: DROP the PENDING "ETA if trend continues" (a forecast wearing a data costume; invites the chase impulse the vetoes exist to block — distance-in-points is honest and sufficient); data_source badge + timestamp/age are LOAD-BEARING — verified first, and the ONLY field allowed to trigger write-side work pre-v1 (a live MTM without a freshness label = stale-as-live). Design: dark-terminal system, one-page layout proposal for sign-off; no design agent.

DEVENDRA — Three PRD edits registered + "All the prompts should be accurate, to the point and professional level"

CLAUDE

Registered as a STANDING STANDARD, operationalized: ACCURATE = every claim evidenced or marked VERIFY (no "likely" where a query can answer); TO THE POINT = exactly four parts — Scope, Bright lines, Deliverables, Acceptance evidence — nothing else, optional items excluded; PROFESSIONAL = enumerated requirements, closed vocabularies, explicit fail behavior per component, and a stop-point. Every remaining prompt in the sequence held to this bar; first application: Monday's P0.1+P0.2 directive.

DEVENDRA — Rule committed to PRD as permanent + "Market is closed"

CLAUDE

Post-close docket ordered: (1) run_postgres.sh fix — diff + dry-run → go → apply (highest recurrence risk); (2) column-width invariant test — must FAIL vs the pre-fix declaration; (3) paper_trade_eq.quantity — diagnose, propose, hold; (4) persistence-audit note; (5) Friday probes: Row K now, Row G through the real code path, Row F after 20:00; (6) the 9-section acceptance query LAST — with preamble sanity count, gap annotation, and the sections read closest: UNMAPPED-by-gate, values_tested_json on POST_CLAMP_RR, the B8 strict check, the 42-char round-trip rows.

DEVENDRA — Item 1 diff + dry-run + NINE SCREENSHOTS + "system generated no signals on good-move days; make F&O accurate, perfect, reliable, professional"

Item 1: find_pgbin() helper checking for the postgres BINARY (not bin/ directory — postgresql-client creates bin/ without the server, exactly the recycle state); 4 dry-run cases (healthy preserved; incident state now fires install; multi-version robust; fresh pod preserved). Screenshots: OI Lab SENSEX (78,202 +1.32%, PCR 1.85, STRONGLY BULLISH 62%, max pain 77,900, R 80,000 / S 77,500); OI Lab BANKNIFTY (58,545 +1.67%, PCR 1.02, MILDLY BULLISH, put writers stepping in); OI Lab NIFTY (24,345 +1.13%, PCR 1.61, STRONGLY BULLISH 64%); NIFTY Option Chain page (BULLISH 60% market read, R1 25,000 / S1 23,000, full chain with OI buildup tags); Trading Desk (ALL F&O indices suppressed banner "MARKET CLOSED ·12, OTHER ·5"; three PLAN 1 OF 1 cards — BASELINE OUTLOOK LONG BIAS, BUY CALL on NIFTY 24,350 /

BANKNIFTY 58,600 / SENSEX 78,200, 45% conf, RANGING, INFO_ONLY, Auto-trade NO, signaled 11:44 IST, exited 13:27 "stale — trigger expired", NIFTY MFE +0.00 MAE –47.98); TradingView 15-min charts of all three indices showing the afternoon breakout through the marked levels.

CLAUDE

Item 1: GO. Then the missed-move diagnosis, reconstructed from the screens: the ANALYSIS layer read the day RIGHT (bullish OI, spot>max-pain, correct direction on all three cards) and the market delivered — the failure is structural, and every cause is already queued: (1) all three setups are BASELINE/INFO_ONLY because the only tradeable lane has no production writer (P0.4-CRITICAL → P1.2); (2) all three signaled at 11:44 — one minute after DB recovery; the incident ate the setup-formation window; (3) all three expired stale at ~13:27, and the market then crossed EVERY trigger (NIFTY 24,288.90→close 24,344; SENSEX 77,989→78,215; BANKNIFTY 58,186→58,558) — trigger-geometry + staleness quantified on a live missed move; a time-expiry with no re-arm structurally cannot catch afternoon breakouts from morning bases; (4) regime said RANGING on a trend day — regime validation is P1.2's test case; (5) the "MARKET CLOSED ·12" suppressions must be timestamped — any in-session firing = live P0.1 evidence. Warned against the trader's chase impulse applied to system design (loosening gates overnight because today got away). Added ONE docket item: capture today as the P1.2 case study (plans, trigger-cross vs expiry times, regime vs realized, suppression timestamps) — Exhibit A for the geometry memo, so the owner's entry-model decision is calibrated against THIS day's numbers.

DEVENDRA — "what should i send to coder now? I want to fix the F&O thing professionally and permanently. 6 months, no result."

CLAUDE

Straight answer first: the work produced a platform that no longer lies to itself, but ZERO TRADES = zero expectancy evidence — the only result a trader can bank; the queue was optimized for per-slice safety while the deliverable that matters sat at the end. Delivered FNO_COMPLETION_MISSION.md (full text in Appendix D): one mission, six phases, one finish line — the first honest, fully-instrumented paper trade within ~10 sessions, then a 20+ signal / 5-session evaluation sample. M0 tonight+weekend (docket + case study + trigger-geometry OPTIONS memo); M1 P0.1+P0.2; M2 drift (unblocks merge, closes with ENV-ISOLATION); M3 PAPER_WRITER discipline + VIX rider; M4 OWNER DECISION GATE (entry model, lifecycle, staleness, regime — scheduled during build so it never blocks); M5 the real TREND_CONTINUATION emitter implementing the owner's decisions, wired to paper execution — first trade, Sites A/B/C acceptance closes; M6 no-touch evaluation sample → expectancy report, the moment the 6–12-month clock officially starts. Checkpoints compressed to one per phase; everything off-path explicitly frozen; failure handling and a 2-session slip rule included. Trader-to-trader note: if M6 says the strategy loses after costs, that is STILL the mission succeeding — knowing is what the next six months get built on.

DEVENDRA — Invariant test run: caught a SECOND live overflow

TRIGGERED_AWAITING_EXECUTION and TRIGGERED_EXPIRED_UNEXECUTED (28 chars) vs execution_status varchar(24); 33 dropped INSERTs since 11:54 IST; zero TRIGGERED-branch rows landed today. Options: (a) emergency ALTER to varchar(32) now; (b) diagnose-only, fix in M3 (~4 days dead); (c) varchar(64)-for-consistency or rename the literals.

CLAUDE

(a) — with one modification: SWEEP FIRST. Precedent identical to the morning; (b) is chosen data loss across two live sessions after the cause is known; (c)'s 64-for-consistency rejected (the invariant test IS the width policy now; columns stay sized to contents+headroom) and renaming production enum vocabulary to fit a column rejected without discussion. Modification: paste the invariant test's full assertion table (every column, max literal vs width, margin) and widen ANY sibling within ~4 chars in the SAME pre-approved pass — one ALTER event, not a third incident Monday. Acceptance addendum: tonight's results now carry TWO annotated gap windows (10:50–11:43 all writes; 11:54→fix TRIGGERED branch), and the B8 strict check returning zero rows must be labeled dropped-writes-not-absent-logic. Noted: Item 2 just paid for the docket — a test written at 17:00 caught, before the weekend, the exact failure class that cost 54 minutes that morning. (Devendra re-sent the same question; Claude restated the ruling compactly: (a) varchar(32), sweep-first, sequence assertion-table → ALTER → Drizzle → green → log-quiet → evidence.)

DEVENDRA — Item 3 diagnosis + Item 4 landed + Item 5 scoping

Item 3 root cause: paper_trade_eq column is qty; three raw-SQL sites in paperAccountReconciliation.ts use "quantity" — a fossil rename that missed string literals; read-only equity-MTM blast radius; 4-occurrence diff proposed with 5-step verification. Item 4: recycle-survival matrix committed. Item 5 asks: Item 3 go/hold; Row K scope (a: 30-symbol sequential 3 req/s; b: 100 symbols; c: one 500-symbol batch call; d: defer a week). M0-A case-study drafting offered in parallel.

CLAUDE

Item 3 (a) GO now — trivially the one-liner as pre-ruled; it also silences a recurring false alarm polluting the error log (2 errors/reconnect) before an unattended weekend, on a platform whose incident tripwire IS that log. Addition: grep "quantity" against ALL paper-equity raw SQL repo-wide — sweep the class, not the instance. Item 5.K: (a)+(c), not (b)/(d) — Row K's consumers are the briefing Checkpoint 0 (breadth feasibility) and M5's contract-selection budget; sequential gives latency/pacing, the single batch call answers wide-universe viability; (b) is the awkward middle. Caveat recorded: off-hours results are a lower-bound; a 60-second in-session spot-check Monday (11:30–14:00, abort on first 429) promotes the row fully. M0-A approved — lead with the suppression-timestamp query (highest-priority element; converts P0.1 to evidence-backed if any fired in-session).

DEVENDRA — Item 3 mechanical-honesty flag: edit landed but api-server runs compiled dist, no hot-reload — two post-edit errors already fired. Options: (a) rebuild+restart now (~60s, post-close); (b) rebuild only, activates at next natural restart; (c) leave source-correct/process-stale.

CLAUDE

(a) rebuild + restart NOW — and praised the flag itself (source-correct/process-stale is exactly how things get "verified" falsely). 19:15 Friday = the single best restart window of the week (maximum distance from Monday 09:15; morning trace already proved clean boot recovery). Cost of NOT restarting: a weekend of false-positive log noise degrading the only unattended monitoring channel, plus a worse property for (b) — a pending deploy that activates whenever the pod happens to die (an uncontrolled deployment time, the day after a recycle incident). Monday must not be this fix's first runtime. Pre-restart check: paste git status — the tree must be clean-intentional (tonight's deltas only: qty rename, invariant test, Drizzle widths). Sequence: status → build → restart → boot trace (Kite reconnect, schedulers, flag survival) → curl MTM → log tail → bug_testing_agent → Item 3 closed on RUNTIME evidence. Acceptance query runs after the restart settles.

DEVENDRA — "what are we doing right now?"

CLAUDE

Orientation: Friday evening mid-docket. Done: Items 1 (self-heal fix), 2 (invariant test + emergency widening after it caught the second overflow), 4 (persistence note), 3 (edited; restart ruled). In flight: rebuild+restart verification chain, Item 5.K sweep+batch, Row F at 20:00, M0-A case study, the acceptance query with two gap windows. Bigger picture: the F&O Completion Mission approved today — tonight is effectively Phase M0; Monday starts M1; the geometry memo lands this weekend for the M4 decision.

DEVENDRA — "are we fixing the f&o thing? how much is done? how much is pending?"

CLAUDE

Trader's scorecard: real F&O paper trades in 6 months = 0 — still true tonight. Done (~60%, the foundation): diagnosis complete (the only tradeable lane never had a writer — proven, not guessed); the truth layer (funnel instrumentation live — nothing dies unexplained anymore); honesty guards (fabrication classes structurally blocked; the invariant test already caught two live bugs); the analysis layer works (today's read was right); today's miss captured as the calibration case. Pending (~40%, the part that produces trades): the mission table M0→M6 — roughly two weeks to the first real trade, three weeks to the first honest answer to "does my strategy make money after costs" — the result six months never produced because a system that never trades can only be rebuilt, not evaluated. The thing to protect: the sequence itself; the owner's only job in it is the M4 decision.

DEVENDRA — "ok can we do the back test with historical data just to save time?"

CLAUDE

Yes — at two specific points, not as a replacement (the platform already learned this: Stage 4's first backtest, 21 trades, net loss, correctly flagged small-sample). (1) M0-B — the big win: the M4 decision is testable on HISTORICAL SPOT DATA ALONE (trigger geometry is a spot-level question): replay 60–90 sessions of 15-min candles per entry model per regime → trigger-hit vs stale rates, MFE/MAE, time-to-trigger — the memo gets a quarter's evidence instead of two days'; potentially saves the 2–3-week wrong-decision redo loop. (2) M5-B — after the real emitter exists, replay it through the Stage 4 engine with provenance labels, PARALLEL to M6. Cannot compress: M5 itself (can't backtest a strategy that doesn't exist); M6 (modeled premiums aren't fills; tuning-to-backtest builds something that only works on the past — forward sample with real premiums is the only proof); no tuning against the backtest mid-sample. Zero new build — warehouse + existing engine.

DEVENDRA — POST-CLOSE DOCKET CLOSED · M0 CHECKPOINT SIGNABLE

All six items landed: Item 1 applied (4 dry-run cases); Item 2 invariant test 10/10 — caught 2 live defects first run, both fixed same-pass (execution_status 24→32, verdict 16→24 sibling); Item 3 qty rename applied + rebuild/restart 19:30 — MTM returns ■15,954.45 (was 0), 0 residual errors, Kite/flag/fixes survived; Item 4 persistence RESOLVED; Item 5 Rows F/G/K all evidenced (Row F participant OI landed 19:31 via same-code-path wrapper — ACTIVE; Row K FEASIBLE off-hours p50=236ms, 0 429s); Item 6 acceptance query ALL ZERO-DEFECT ASSERTIONS PASS on today's 91-row sample, 3 gap windows annotated, Step 2 closes on Fri+Mon combined. M0-A case study written as Exhibit A (P0.1 evidence flagged UNRESOLVED — suppression counters in-memory only). Three independent bug_testing_agent verdicts, all fixed. Two live P0s discovered by tonight's tools and closed same-night. Next: Monday M1 + suppression-persistence rider; Monday Row K re-probe; M0-B memo; owner reads the case study.

CLAUDE

M0 CHECKPOINT: SIGNED. Sign-off notes: (1) the weekend deliverable is the HISTORICAL STUDY + memo together — M0-B replays the three entry-model candidates over 60–90 sessions of warehouse spot candles, descriptive statistics per option per regime, NO ranking (ranking is the M4 decision); warehouse coverage verified first so claimed sample = real sample. (2) Suppression-persistence as M1 rider approved — from Monday, every suppression event gets a timestamped row and convicts itself. (3) Monday Row K re-probe approved with guardrails (11:30–14:00, 60s, 3 req/s, abort on first 429). (4) Devendra's weekend reading = the case study — the M4 decision derived from it is the single highest-leverage trading call in the mission. Week verdict: two live incidents caught and killed same-day, funnel instrumented and verified, mission signed and fueled.

DEVENDRA — Four sign-off notes committed to PRD; weekend ledger closed (noting M0-B gated on coverage check — "this pod's candle tables were empty in Row A audit")

CLAUDE

Pre-ruled the buried blocker: if the warehouse is truly empty, M0-B has no data. Weekend path pre-authorized: Step 1 coverage check (prod vs dev DB — Row A may have hit the wrong one); Step 2 if empty — a BOUNDED BACKFILL pre-approved (~7,000 rows of 15-min OHLC) with three conditions: (a) check Kite historical-API ENTITLEMENT first — it's a paid add-on; if absent, stop and report cost (no Yahoo candles feeding a decision memo); (b) provenance labels source='KITE_HISTORICAL_BACKFILL' + fetch timestamp; (c) rate-respectful, off-hours. Step 3 fallback: reduced study on trusted daily candles, honestly labeled. Flagged for Devendra: if the entitlement is missing, the add-on is likely the best money the mission spends — his call.

DEVENDRA — KITE HISTORICAL API: ENTITLED (probe: 100 15-min NIFTY candles in 386ms, volume=0 on cash-index as designed). Backfill sized: 84 REST calls, ~2–3 min off-hours, provenance-labeled. Weekend closed: owner reads case study; agent runs backfill → replay 3 models × N staleness windows × 90 sessions × 3 indices → stats tables → ≤2-page memo; Monday M1 + Row K spot-check. "Good weekend. ■"

CLAUDE

Week wrap: opened with a phantom demoter; closes with the funnel instrumented and verified, five live defects found and killed by evidence (test-file prod leak, two varchar overflows, a fossil rename, a self-heal check bug), two incidents recovered inside emergency lanes that held, the mission signed, and the M4 decision about to be calibrated against 90 sessions of real candles instead of a hunch — more forward motion on the F&O core than any prior month, because every step ran on evidence. Devendra's homework: read the case study AS THE TRADER — the memo's question is already visible in the numbers: where should the entry have been, and how long should the trigger have lived? Next contact: memo → M4 decision → Monday M1. ■

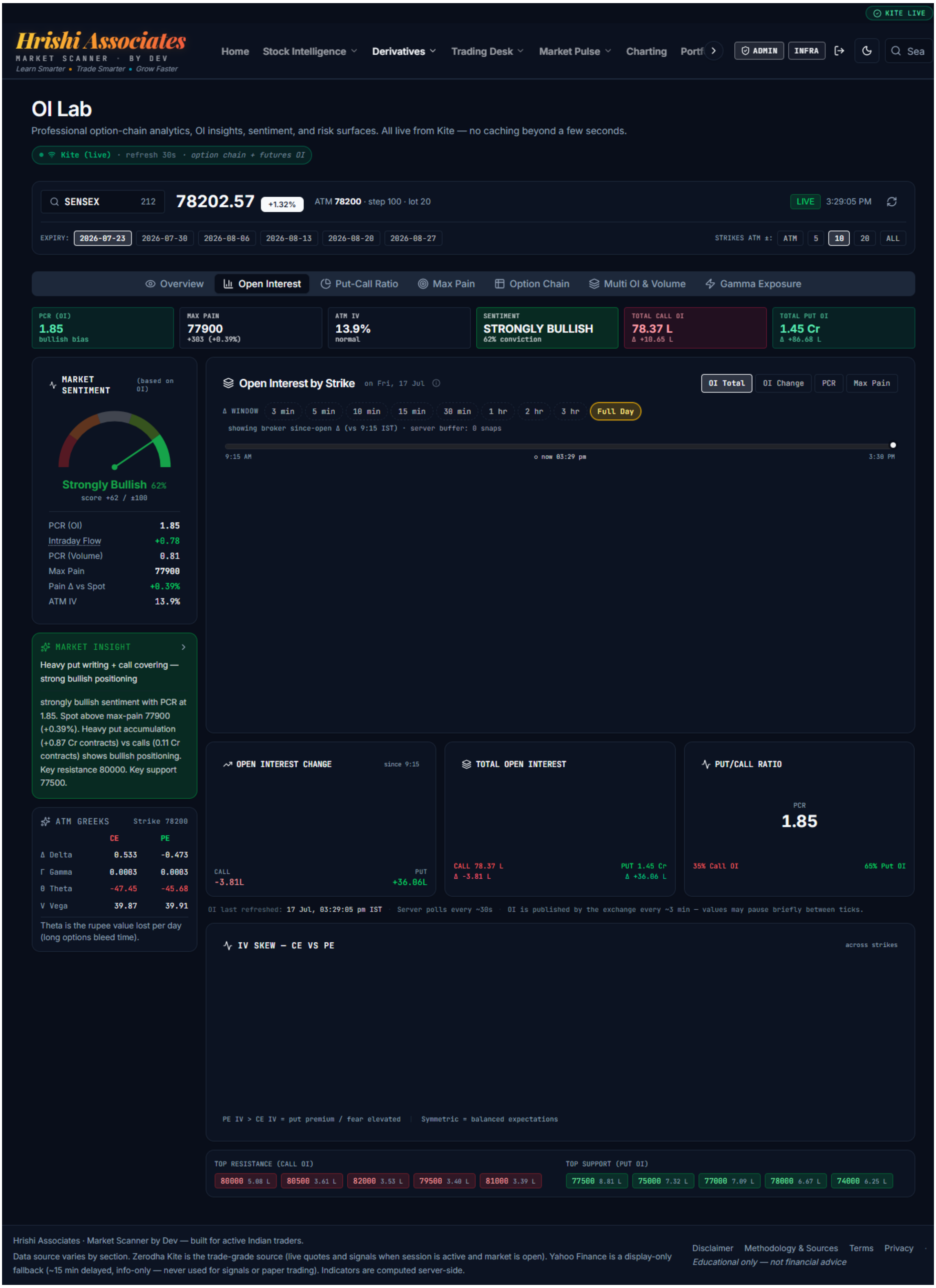
DEVENDRA — "give a pdf for the entire chat. Do not miss anything. Include everything… from starting to end."

CLAUDE

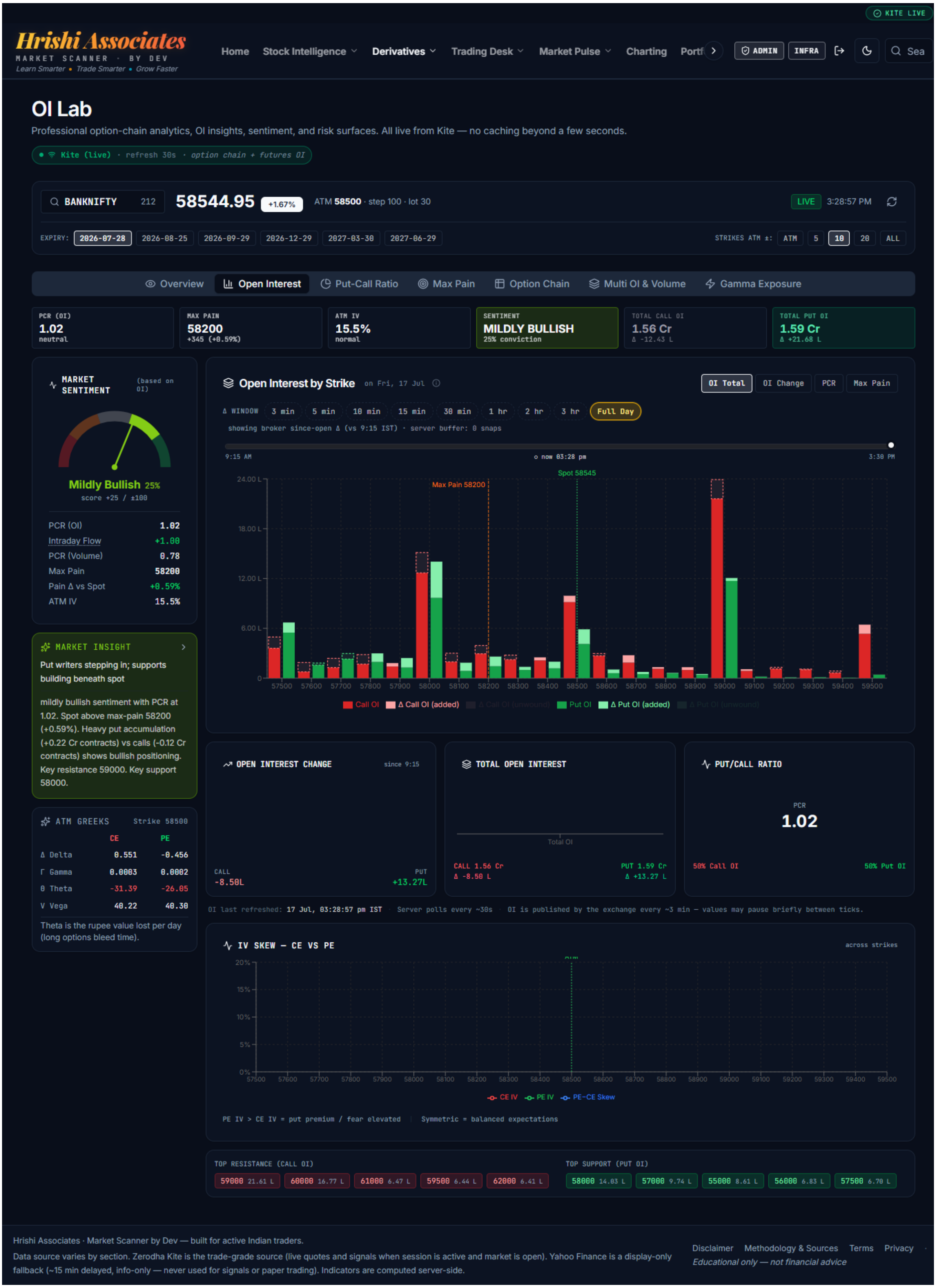
Produced this document: the complete turn-by-turn record with all nine screenshots embedded and the four source documents reproduced in full as Appendices A–D.

PART 6 — THE NINE SCREENSHOTS (UPLOADED 17 JULY 2026)

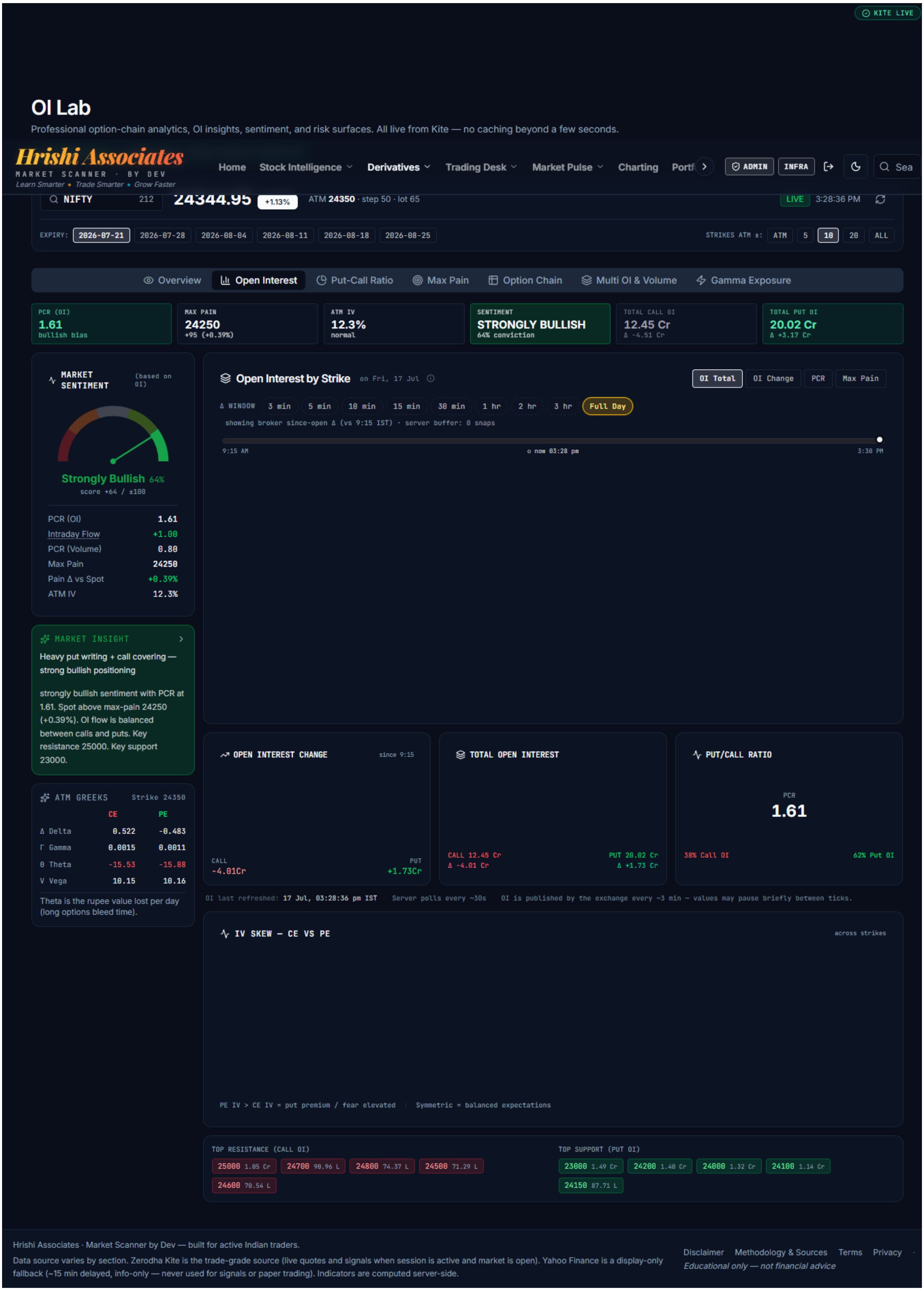
Screenshot 1 — OI Lab · SENSEX (15:29 IST): 78,202.57 +1.32% · PCR 1.85 · STRONGLY BULLISH 62% · Max Pain 77,900



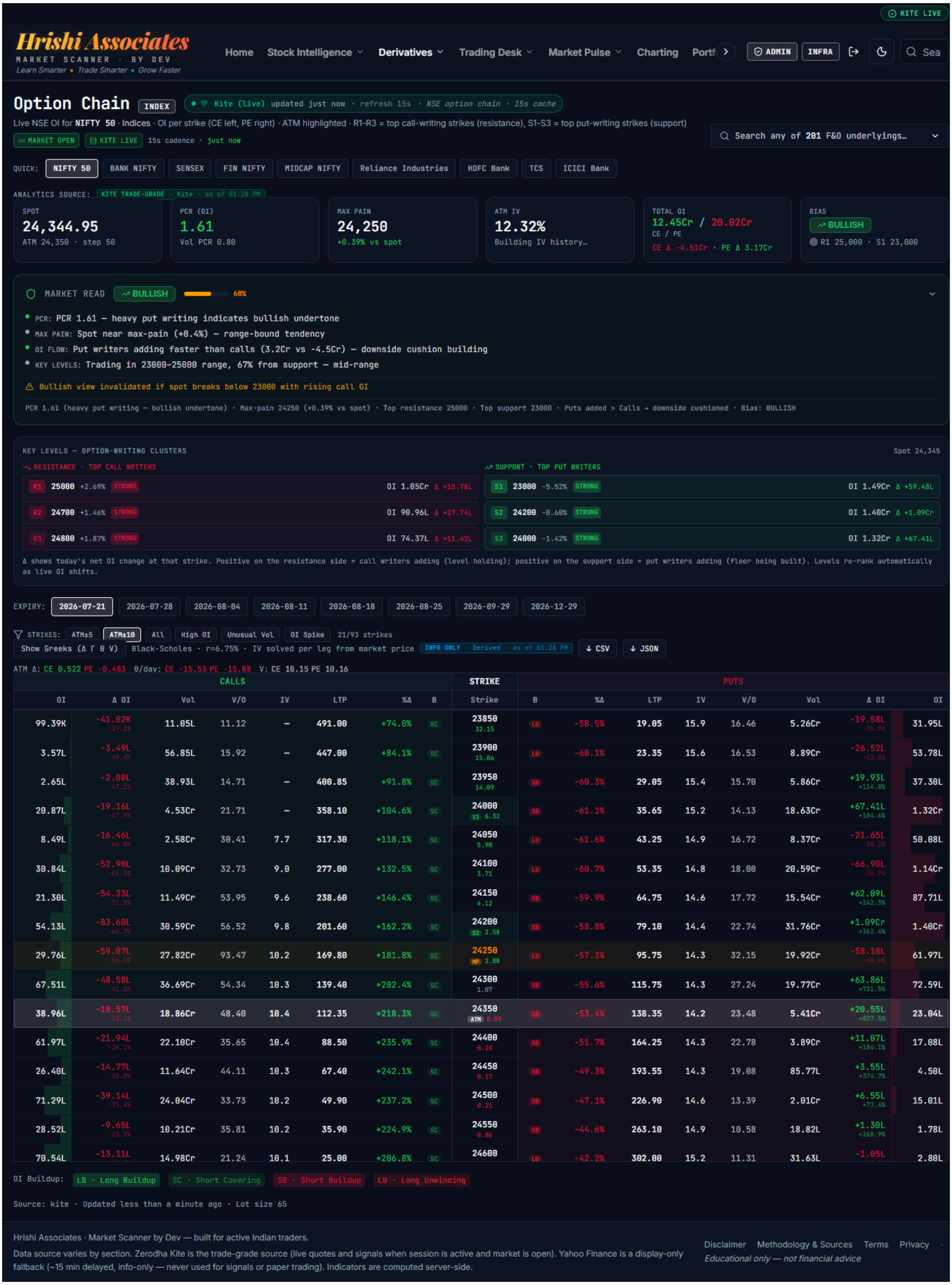
Screenshot 2 — OI Lab · BANKNIFTY (15:28 IST): 58,544.95 +1.67% · PCR 1.02 · MILDLY BULLISH · put writers stepping in



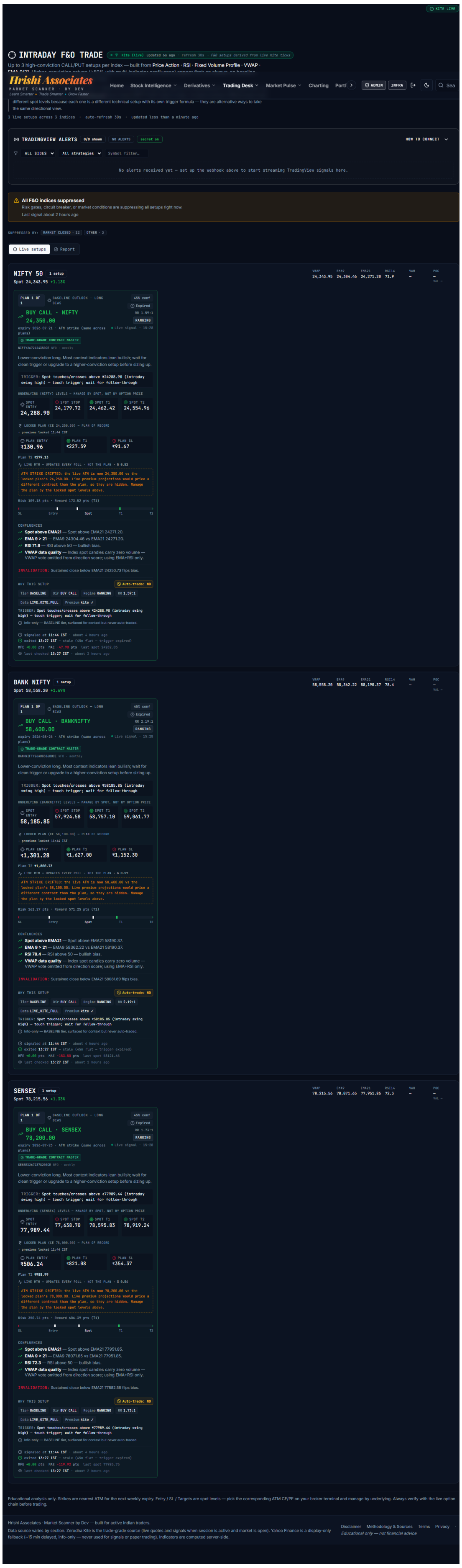
Screenshot 3 — OI Lab · NIFTY (15:28 IST): 24,344.95 +1.13% · PCR 1.61 · STRONGLY BULLISH 64%



Screenshot 4 — Option Chain page · NIFTY 50: Market Read BULLISH 60% · R1 25,000 / S1 23,000 · full chain with OI buildup tags



Screenshot 5 — Trading Desk · Intraday F&O Trade: all indices suppressed banner; three BASELINE INFO_ONLY plans (signaled 11:44, expired stale 13:27)



Screenshot 6 — TradingView · SENSEX 15-min (15:25 IST): breakout through 78,033 marked level

Screenshot 7 — TradingView · SENSEX 15-min (16:51 IST): close 78,256 near highs

Screenshot 8 — TradingView · BANKNIFTY 15-min (16:50 IST): close 58,576, cleared 58,400

Screenshot 9 — TradingView · NIFTY 15-min (16:50 IST): close 24,343.65, reclaimed 24,300

APPENDIX A — FNO_SWING_MASTER_FIX_PROMPT_2026-07-15.txt (uploaded by owner, verbatim)

```
# MASTER CODER PROMPT — PROFESSIONAL-GRADE F&O INTRADAY + SWING CASH ENGINE AUDIT, REPAIR, VALIDATION AND PRODUCTION
HARDENING

**Project:** Devendra Sharma’s Indian Market Scanner / Market Intelligence and Trading Decision Platform
**Primary objective:** Repair and harden the complete F&O intraday and Swing Cash engines—from trusted data ingestion
through signal generation, paper-trade execution, entry, stop-loss, targets, exits, P&L, reports, alerts, diagnostics
and production verification.
**Operating mode:** Paper trading and decision support only. Do not enable live broker order placement unless
separately and explicitly approved.
**Time standard:** Every market clock, timestamp, scheduler, report, expiry calculation and user-facing time must use
`Asia/Kolkata` / IST.

---

## 1. YOUR ROLE

Act simultaneously as:

1. A senior TypeScript/React/Express/PostgreSQL full-stack engineer.
2. A professional Indian index-options and cash-market Swing trader.
3. A quantitative strategy auditor.
4. A market-data reliability engineer.
5. A production incident investigator.
6. A risk manager responsible for preventing false, duplicated, stale, contradictory or financially misleading trades.

This is not a cosmetic UI task. This is a source-code, database, market-data, trading-logic, execution-simulation,
risk-control and reporting audit-and-fix assignment.

Do not claim that an engine is “perfect,” “professional,” “high conviction,” “production ready,” or “profitable”
merely because code compiles or the website loads. Every claim must be supported by code-path evidence, data
reconciliation, test results and paper-trade outcomes.

---

## 2. FINAL PRODUCT STANDARD

The platform must operate as an **Indian Market Intelligence and Trading Decision Platform**, not as a collection of
unrelated indicators.

The mandatory end-to-end loop is:

`Trusted market data → normalization → freshness/source validation → market regime → directional bias → setup
detection → signal gates → strike/candidate selection → immutable trade plan → paper execution → live lifecycle
monitoring → exit → realistic net P&L → attribution → reports → expectancy feedback`

There must be:

- One source of truth for every price, signal, direction, trade plan and P&L number.
- No fabricated data.
- No stale data silently treated as live.
- No delayed fallback used for F&O entry or exit.
- No random BUY/SELL alerts.
- No silent signal rejection.
- No hidden execution failure.
- No server/UI disagreement.
- No gross P&L presented as net P&L.
- No option P&L calculated from spot-index movement.
- No Swing recommendation that contradicts its score without an explicit reason.
- No production status declared without evidence.

---

## 3. PRESERVE THE STRENGTHS ALREADY BUILT

Do not destroy or casually rewrite verified working areas. Preserve and regression-test:

- Central/trusted data-layer architecture.
- Kite-first live market-data policy.
- Source, freshness and trust labels.
- Data-honesty behavior when data is missing.
- Owner-only diagnostic routes and security protections.
- Kite token encryption and safe export handling.
- Instrument-master based symbol resolution.
- Portfolio and chart source-honesty fixes.
- Option-chain trusted-source migration.
- F&O signal tiers:
  - `TRADEABLE_SIGNAL`
  - `WATCHLIST_SETUP`
  - `INFO_ONLY`
- The rule that recovery-veto/chase-veto conditions demote to `INFO_ONLY` rather than silently fabricating or forcing
trades.
- Existing anti-flip cooldown and circuit breakers unless evidence proves a defect.
```

- F&O signal-plan immutability and audit history.
- OI unit handling as contracts, not lots, unless direct Kite documentation/data reconciliation disproves it.
- Gross-vs-net display separation.
- Option exit-premium market-shadow columns and provenance.
- Telegram owner alerts for qualified F&O signals and Swing staged-order lifecycle.
- Pre-market and post-market daily Telegram reports.
- Existing database constraints, risk caps and paper-only safety gates.
- Existing working tabs/features. Repair them; do not remove them to make tests pass.

Any proposed change to a currently verified risk rule must be isolated, justified with evidence, backtested, shadow-tested, owner-approved and reversible.

---

4. KNOWN CRITICAL BUGS AND OPEN RISKS — TREAT THESE AS MANDATORY

Audit and close each item. Do not mark it complete based only on code inspection.

4.1 False “Market is closed” on `/options`

The F&O Intraday page has falsely shown “Market is closed” during normal IST market hours because of stale browser/React Query cache and a deprecated fallback such as `marketState="closed"`.

Mandatory rule:

- Only `marketStatus.marketOpen === false` from a current trusted response may render the market-closed state.
- Missing status, stale status, loading, API error or schema mismatch must render a neutral loading/degraded/error state—not “market closed.”
- Normal NSE session defaults to 09:15–15:30 IST unless the official holiday/special-session configuration says otherwise.
- Remove or quarantine every deprecated “closed by default” fallback.
- Revalidate cache keys, stale times, refetch-on-focus, refetch interval, invalidation and hydration behavior.
- Add tests for 09:14, 09:15, 12:00, 15:29, 15:30, holiday, special session, missing API response and stale cached response.

4.2 `/api/options/signals` HTTP 500 and schema drift

A previous failure came from `contractInstrumentToken` and Zod/OpenAPI/API response mismatch.

Mandatory outcome:

- `/api/options/signals` returns HTTP 200 for valid degraded/no-signal states.
- Optional/nullable fields are represented consistently across:
  - database schema
  - server types
  - Zod schema
  - OpenAPI
  - generated client
  - frontend TypeScript types
  - tests
- Response must include rich `marketStatus`, `setupState`, `dataQuality`, `warnings`, `blockingReasons` and signal/execution truth.
- Do not hide a schema failure by stripping fields.

4.3 Execution truth stripped by OpenAPI/Zod

The `execution` object was previously removed from signal-history/API output, causing the UI to show false `NOT_CONFIRMED` popups.
Mandatory rule:

- Preserve `execution` end-to-end.
- Distinguish:
  - `INFO_ONLY`: no paper trade is expected.
  - `WATCHLIST_SETUP`: trigger is pending.
  - `TRADEABLE_SIGNAL`: eligible for paper execution after all gates.
  - `EXECUTION_CONFIRMED`: paper position created.
  - `EXECUTION_REJECTED`: eligible signal failed a specific execution gate.
  - `EXECUTION_ERROR`: technical failure.
- Never label an `INFO_ONLY` row as failed paper execution.
- Every tradeable signal must have a deterministic execution outcome and reason.

4.4 Kite session/pre-open fragility

A known no-signal day occurred because the Kite session expired before market open and login happened late.

Mandatory fixes:

- Pre-open Kite/session readiness check before 09:15 IST.
- Clear owner alert when session is invalid, market-data subscription is unavailable or instrument sync failed.
- Global degraded-data banner.
- F&O-specific no-live-data state and skip reason.
- Do not generate a paper trade from fallback/delayed data.
- Persist session/data-health events for later audit.
- Provide a pre-open readiness checklist and production run evidence.

4.5 Data infrastructure reliability

Audit and repair:

- Candle warehouse SQL failure.
- Safe historical-candle backfill.
- Full NSE scan timeout that returns a cached/partial universe such as 198 rows.
- NSE bhavcopy failures.
- Yahoo or any untrusted fallback leakage.
- Kite `ECONNRESET`, reconnect, resubscribe and gap recovery.
- Stale TradingView alerts.
- Scheduler duplication and missed runs.
- Process-local latches that reset on restart.

The system must fail honestly and visibly. Partial/cached results require explicit provenance, coverage and as-of fields.

4.6 F&O valid candidates converting to zero opens

Trace every stage:

`candidate → signal → tier → gate → option contract → paper sizing → execution → open trade`

Build a complete skip-reason dashboard and persisted diagnostics. Determine whether zero opens are caused by correct safety rules, data failure, schema loss, stale inputs, sizing, duplicate protection, contract resolution or an actual strategy-quality issue.

4.7 Swing scanner reliability

Audit and repair:

- Symbol-resolution errors.
- Inactive/renamed securities.
- NIFTY 500 universe completeness.
- Sector/industry mapping coverage.
- Missing candles and benchmark data.
- Scan timeouts and partial results.
- High-score candidates rejected without visible reasons.
- UI label versus execution-label mismatch.
- Whether sector/relative-strength fields are used in scoring or merely displayed.

4.8 Backtest honesty
The earlier option backtest used synthetic assumptions such as fixed spot percentage, fixed delta and no genuine theta/IV behavior.

Mandatory provenance labels:

- `REAL_CHAIN`
- `BLACK_SCHOLES_MODELLED`
- `SYNTHETIC_DELTA_PROXY`

Never merge these into one performance number without a breakdown. Real paper-trade results and synthetic/modelled results must remain separately identifiable.

---

5. EXECUTION PROTOCOL — DO NOT ATTEMPT AN UNCONTROLLED FULL REWRITE

This is a master scope, but it must be executed in bounded phases.

For every phase:

1. Inspect the actual current code and database.
2. State what is implemented, partially implemented, dead/reference-only or missing.
3. Produce a file/function/data-path map.
4. List defects with severity and evidence.
5. Make the smallest safe root-cause fix.
6. Add/update tests.
7. Run typecheck, unit tests, integration tests and manual smoke tests.
8. Record exact files/functions/migrations changed.
9. Provide before/after proof.
10. State residual risks.
11. Do not start an unrelated lane.

Do not repeatedly loop through vague “planning next move” activity. Maintain a checklist with stable task IDs and mark each item `NOT_STARTED`, `IN_PROGRESS`, `BLOCKED`, `DEV_VERIFIED`, `PROD_VERIFIED` or `LIVE_SAMPLE_PENDING`.

PART A — COMMON DATA, TIME, STATE AND RISK BACKBONE

6. BASELINE AND CODE MAP

Before changing strategy logic, create:

`FNO_SWING_BASELINE_AND_CODE_MAP.md`

It must include:

- Repo/monorepo structure.
- Every server route used by F&O and Swing.
- Every core library/function used by:

  - market data
  - candles
  - instrument resolution
  - signal generation
  - paper execution
  - lifecycle
  - exits
  - P&L
  - reports
  - alerts
- All database tables and relationships.
- All schedulers/background workers.
- All in-memory state/latches.
- All frontend pages and API hooks.
- All tests currently covering these paths.
- Implemented logic versus reference/example/dead code.
- Production code versus development-only code.
- Current environment flags that change behavior.
Run and record:

```bash
pnpm install
pnpm run typecheck
pnpm --filter @workspace/api-spec run codegen
pnpm --filter @workspace/api-server run test
pnpm --filter @workspace/scanner run test
pnpm run typecheck
```

Also run focused tests for database migrations, signal engines, cost model, execution lifecycle, timezone and API contracts.

---

7. TRUSTED DATA POLICY

Create one canonical `DataEnvelope<T>` or equivalent for all decision-critical data:

```ts
type DataEnvelope<T> = {
  data: T | null;
  source: string;
  sourceClass: "TRUSTED_LIVE" | "TRUSTED_EOD" | "MODELLED" | "UNTRUSTED_INFO_ONLY";
  asOf: string | null;
  receivedAt: string;
  ageMs: number | null;
  fresh: boolean;
  complete: boolean;
  coverage?: number;
  warnings: string[];
  notForTradeDecisions: boolean;
};
```

Mandatory rules:

- Kite is the priority source for live prices, instruments, tradable contracts and decision-critical intraday candles.
- Yahoo or delayed public sources must not feed F&O entry, exit, stop, target or MTM.
- Supplementary fundamentals/news/corporate data must be source-labelled and must not silently substitute for live price data.
- Missing data remains `null`; do not create plausible-looking constants.
- Every API response and UI card must expose source and as-of information.
- Every signal record must snapshot the exact data provenance used.
- Every paper exit must record the premium source and freshness.
- Any stale or untrusted dependency must either block the trade or demote it to `INFO_ONLY`, according to a documented rule.
- A failure to obtain data must never be translated into bearish, bullish or market-closed status.

Create `GET /api/fno/data-health` and `GET /api/swing/data-health` with status, freshness, coverage, warnings and blocking reasons.

---

8. INSTRUMENT MASTER AND SYMBOL RESOLUTION

Maintain a canonical daily instrument master with:

- exchange
- segment
- tradingsymbol
- underlying
- instrument token
- contract instrument token
- expiry
- strike
- option type
- lot size
- tick size

- active flag
- first/last seen
- renamed predecessor/successor where known
- sector and industry for equities
- benchmark/universe membership

Rules:

- Do not resolve by fragile string concatenation where instrument-master lookup is possible.
- NFO/BFO and NIFTY/BANKNIFTY/SENSEX mappings must be explicit and tested.
- Current and next expiry must be derived from the active master, not assumptions.
- Strike step must come from verified contract metadata/config.
- Every unresolved symbol/contract must generate a persisted diagnostic reason.
- Renamed/inactive Swing stocks must be handled without crashing or silently shrinking the universe.

---

9. TIMEZONE, MARKET CALENDAR AND SCHEDULERS

All code must use IST-aware functions.

Audit:

- market-open checks
- candle boundaries
- option expiry dates
- daily reset
- weekly reset
- forced exit
- report generation
- pre-market and post-market jobs
- query date filters
- database timestamps
- frontend formatting

Rules:

- Store timestamps in UTC where appropriate, but convert through an explicit IST market-calendar service.
- No server-local timezone assumption.
- No `new Date().getHours()` market logic without timezone conversion.
- Normal session: 09:15–15:30 IST, overridden only by official holiday/special-session config.
- Persist scheduler runs and dedup keys.
- Restarts and multiple replicas must not duplicate signals, trades, reports or alerts.

---

10. PERSISTENT STATE AND RESTART SAFETY

Identify every module-level variable used for:

- signal deduplication
- anti-flip cooldown
- stop cooldown
- max trades
- daily/weekly loss caps
- forced exit
- report deduplication
- scan latches
- scheduler completion
- baseline lane locks
- missed-signal buffers
- cold-start reconciliation

Move decision-critical state to PostgreSQL or protected transactional/advisory-lock state.

Acceptance tests:

- Restart during market hours.
- Restart with open F&O trade.
- Restart with open Swing trade.
- Restart before forced exit.
- Two application instances receiving the same tick/signal.
- Duplicate scheduler delivery.
- Delayed webhook replay.
- Database reconnect.

No duplicated paper trade, missed exit, reset risk cap or duplicate Telegram alert is acceptable.

PART B — F&O INTRADAY ENGINE

11. CURRENT F&O UNIVERSE

Treat `NIFTY`, `BANKNIFTY` and `SENSEX` as the currently accepted production universe unless direct code and active configuration prove otherwise.

Do not activate FINNIFTY, MIDCPNIFTY, NIFTYNXT50 or stock options merely because old design documents mention them.

Expansion requires:

- active contract support
- data coverage
- separate backtest
- liquidity rules
- risk limits
- owner approval

---

12. F&O END-TO-END PIPELINE

Map and make auditable:

`Index quote → intraday candles → context → regime → bias → setup → tier → option-chain confirmation → contract selection → premium snapshot → immutable plan → sizing → execution gate → paper open → tick/candle monitoring → exit → charges → net P&L → report`

Every stage must output:

- input snapshot IDs/timestamps
- result
- score/confidence
- warnings
- gate decision
- machine-readable reason codes
- human-readable explanation

No stage may silently return `null` without a diagnostic.

---

13. MARKET CONTEXT AND REGIME ENGINE

Classify each index at minimum into:

- `BULLISH_TREND`
- `BEARISH_TREND`
- `RANGE`
- `BREAKOUT_ATTEMPT`
- `VOLATILE_EVENT`
- `GAP_UP_CONTINUATION`
- `GAP_UP_EXHAUSTION`
- `GAP_DOWN_CONTINUATION`
- `GAP_DOWN_REVERSAL`
- `EXPIRY_DECAY`
- `LOW_LIQUIDITY_OR_BAD_DATA`
- `NO_TRADE`

Audit actual inputs and formulas:
- higher-timeframe trend
- EMA structure
- VWAP and VWAP slope
- opening range
- previous-day high/low
- CPR/pivots if used
- ATR and realized volatility
- volume and relative volume
- breadth
- India VIX
- futures price/OI if available
- option-chain structure
- gap percentage and gap fill
- time of day
- expiry proximity
- scheduled event risk

Mandatory principles:

- Regime must be deterministic for identical inputs.
- Missing input must reduce confidence or create a degraded state; it must not default to a confident regime.
- Trend setups cannot be approved in a confirmed range unless their own breakout transition is validated.
- Mean reversion cannot be approved in a strong trend without a documented exception.
- Expiry/event rules must be explicit.
- Store regime changes through the day for later trade attribution.

---

14. SINGLE-SOURCE DIRECTIONAL BIAS

There must be one canonical bias function used by:

- F&O cards
- signal generator
- paper trader
- alerts
- reports

- diagnostics

Output:

```ts
type BiasResult = {
  side: "BULLISH" | "BEARISH" | "NEUTRAL";
  rawScore: number;
  normalizedScore: number;
  confidence: number;
  htfTrend: string;
  htfVeto: boolean;
  components: Record<string, number | string | boolean | null>;
  contradictions: string[];
  dataQuality: string;
};
```

Audit for any independent/legacy bullish-bearish path. It must not be able to create paper trades outside the canonical bias pipeline.

Verify the earlier regression condition where cards and paper trade direction disagreed. Add tests proving alignment.

---

15. F&O SETUP ENGINES

Audit the actual production setup list, expected to include:

- `TREND_CONTINUATION`
- `VWAP_RECLAIM`
- `VOLUME_BREAKOUT`
- `EMA_PULLBACK`
- `MEAN_REVERSION`
- `BASELINE`

Also find any legacy/hidden setup.

For every setup, document and test:

1. Market regime eligibility.
2. Bullish rule.
3. Bearish rule.
4. Required candle timeframe.
5. Trigger transition—not just current state.
6. Confirmation period.
7. Volume requirement.
8. Bias alignment.
9. HTF veto behavior.
10. Recovery/chase-veto behavior.
11. Invalidation.
12. Re-entry/cooldown.
13. Duplicate suppression.
14. Time-of-day eligibility.
15. Expiry/event handling.
16. Confidence formula.
17. Minimum data quality.
18. Expected signal tier.
19. Exact skip reasons.
20. Historical expectancy by index/regime/time bucket.

Do not treat declared-but-null setup directors as implemented. Classify them truthfully.

15.1 Setup-specific target design

Do not blindly hardcode new numeric thresholds. First map current values, then validate via walk-forward testing and configure approved values.

Trend continuation
- Requires aligned HTF/intraday trend, price acceptance beyond structure, supportive VWAP/EMA behavior and no late chase.
- Entry only after a trigger candle closes/holds; not merely because trend is bullish/bearish.
- Invalidate on structure failure or opposite bias transition.

VWAP reclaim/loss
- Must be a state transition: prior position on one side, reclaim/loss, acceptance/confirmation and no immediate rejection.
- Avoid repeated firing while price remains on the same side.
- Include VWAP slope, distance and volume context.

Volume breakout
- Requires defined range/level, valid breakout close, relative-volume expansion, acceptable wick/rejection behavior and follow-through rule.
- Must distinguish real breakout from opening volatility noise.

EMA pullback
- Requires established trend, controlled retracement, defined EMA zone, bullish/bearish rejection and resumption trigger.

- Do not fire from a random EMA touch.

Mean reversion
- Only in range/overextension regimes.
- Requires defined range edge/overextension, failed continuation and reversal evidence.
- Hard-block or demote in strong trend unless separately validated.

Baseline
- Must remain a lower-conviction lane.
- Must never be presented as equivalent to a tradeable high-conviction setup.
- Preserve separate sizing, confidence floor and risk restrictions.
- `INFO_ONLY` baseline outcomes must not be shown as execution failures.

---
16. SIGNAL TIERS AND GATES

Canonical tiers:

- `INFO_ONLY`
- `WATCHLIST_SETUP`
- `TRADEABLE_SIGNAL`

Canonical final decisions:

- `EXECUTABLE`
- `WATCH`
- `DEMOTED`
- `REJECTED`
- `SHADOW_FAIL`
- `DATA_BLOCKED`

Layered gates:

1. Market calendar/session.
2. Data quality/freshness.
3. Direction.
4. Setup transition.
5. Regime compatibility.
6. HTF veto.
7. Recovery/chase veto.
8. Anti-flip/cooldown.
9. Option-chain confirmation.
10. Contract availability.
11. Liquidity/spread.
12. Risk-reward.
13. Daily/weekly risk.
14. Portfolio heat/open-trade cap.
15. Duplicate/dedup.
16. Time cutoff.
17. Execution creation.

Every gate must store:

- pass/fail/demote
- reason code
- values tested
- threshold/config version
- timestamp

Do not use one score to conceal a failed mandatory gate.

---

17. OPTION-CHAIN AND DERIVATIVE CONFIRMATION

Use the trusted option-chain facade. Audit:

- expiry selection
- spot source
- ATM calculation
- strike coverage
- CE/PE OI
- change in OI
- volume
- IV
- bid/ask
- spread
- PCR variants
- max pain
- ATM straddle
- expected move
- support/resistance walls
- writing/unwinding classification
- missing strike detection
- stale snapshot behavior
- OI units

Initial implementation rule:

- Keep new option-chain logic in shadow mode until enough live samples are reviewed.
- Record whether it confirms, contradicts or is unavailable.
- Do not silently turn shadow logic into a hard gate.
- After evidence review, promote only approved fields/gates.

For a CE directional trade, evaluate at minimum:

- bullish spot/futures structure
- put-side support behavior
- call resistance distance
- chosen CE liquidity
- spread
- IV/premium behavior
- expiry risk

For a PE directional trade, evaluate the mirror conditions.

Do not oversimplify PCR or max pain into a direct BUY/SELL signal.

---

18. CONTRACT AND STRIKE SELECTION

Create one canonical selector.

Inputs:

- underlying/index
- side
- spot
- expiry
- time to expiry
- strike step
- target delta/moneyness policy
- premium range
- OI
- volume
- bid/ask spread
- depth if available
- tick size
- lot size
- IV
- stale/missing flags

Principles:

- Liquidity first.
- Prefer ATM or slightly ITM/approved moneyness; avoid arbitrary far OTM contracts.
- Expiry-aware behavior.
- No contract without valid instrument token.
- No contract with missing premium/bid/ask where required.
- No contract with unacceptable spread or insufficient activity.
- Store all considered candidates and rejection reasons.
- Snapshot selected contract metadata into the immutable signal plan.
- Do not retroactively change strike/expiry/entry plan after emission.

Test NIFTY/NFO, BANKNIFTY/NFO and SENSEX/BFO separately.

---

19. IMMUTABLE F&O SIGNAL PLAN

When a tradeable signal is emitted, persist an immutable plan containing:

- signal ID
- index
- side
- setup
- regime
- tier
- confidence/grade
- option symbol/token
- expiry
- strike
- CE/PE
- lot size
- planned quantity/lots
- spot trigger
- option trigger/entry model
- spot invalidation
- option stop
- targets
- trailing rules
- time stop
- force-exit time
- data sources/as-of
- strategy/config version
- cost model version

- all gate results
- creation timestamp

Any later recalculation must be stored as evaluation/audit history, not overwrite the original plan.

Add database constraints/tests proving plan immutability.

---

20. F&O ENTRY RULES

A paper entry is permitted only when all required conditions are true:

- Market is open according to the authoritative IST status.
- Trusted live data is fresh and complete enough.
- Canonical bias is non-neutral.
- Setup transition is confirmed.
- Regime supports the setup.
- Required vetoes/cooldowns permit it.
- Contract is active and resolved.
- Option premium is fresh.
- Liquidity/spread rules pass.
- Planned risk-reward passes.
- Risk/sizing gates pass.
- No duplicate trade or conflicting open position.
- Entry is before the configured cutoff.
- Immutable plan is stored.
- Paper execution record is created transactionally.

Entry price hierarchy must be explicit, for example:

1. Actual qualifying option tick/quote at or after trigger.
2. Executable ask/mid/last policy approved for paper simulation.
3. Never use spot price as option entry.
4. Never backfill an unrealistically favorable premium after the fact.

Record:

- decision timestamp
- signal timestamp
- quote timestamp
- simulated order timestamp
- simulated fill timestamp
- bid/ask/last
- assumed slippage
- fill rule
- latency
- reason if not filled

Distinguish signal generation from actual paper fill.

---

21. F&O STOP-LOSS, TARGET AND EXIT RULES

The engine must use both:

- **Spot thesis/invalidation levels**, and
- **Option premium execution levels**

They serve different purposes and must not be confused.

21.1 Initial stop

Audit current logic and implement a documented hierarchy:

- Spot structure invalidation: breakout failure, VWAP loss/reclaim failure, swing level, ATR-based invalidation or setup-specific level.
- Option premium stop: derived from executable option entry and approved loss/risk model.
- Sanity caps: reject a trade if the required stop is too tight, too wide, below tick-size practicality or creates invalid reward-to-risk.

Whichever approved stop condition triggers first must create a deterministic exit event with source and timestamp.

21.2 Targets

Targets must be setup- and volatility-aware, not arbitrary percentages.

Design and validate:

- `T1`: first risk multiple or nearby structure objective.
- `T2`: continuation objective.
- Optional runner/`T3` only with evidence.
- Store both spot objective and expected/actual option premium objective.
- Do not promise a fixed option premium target based solely on spot ATR.

21.3 Partial booking and trailing
Partial booking/T1 trail is a strategy change. Implement in shadow/design mode first unless already approved.

Proposed auditable lifecycle:

- On T1: book configured partial quantity.
- Move remaining stop to breakeven or approved trail only if rules say so.
- Trail using structure/VWAP/EMA/ATR/high-low mechanism.
- Record each partial fill, remaining quantity and realized/unrealized P&L.

No hidden averaging, martingale or widening stop after entry.

21.4 Mandatory exits

Audit and test:

- option stop
- spot invalidation
- T1/T2/T3
- trailing stop
- opposite confirmed signal
- data-loss safety exit
- expiry-related exit
- time stop
- configured intraday forced exit, including current 15:20 behavior if that is the verified rule
- end-of-session cleanup
- manual owner close
- circuit-breaker close

A missing live premium at exit must not fabricate a favorable value. Use a clearly defined fallback hierarchy and mark the result with provenance/quality.

---

22. F&O POSITION SIZING AND RISK

Create one canonical risk engine.

Inputs:

- paper account equity
- available capital
- risk per trade
- option entry
- option stop
- lot size
- open positions
- index concentration
- daily realized P&L
- weekly P&L
- drawdown
- portfolio heat
- signal tier/setup
- expiry/event risk

Rules to audit/configure:

- Max loss per trade.
- Max lots.
- Max concurrent F&O positions.
- Max one-sided correlated exposure.
- Max daily loss.
- Max weekly loss.
- Drawdown latch.
- Heat cap.
- Baseline lane sizing.
- Post-stop cooldown.
- Anti-flip cooldown.
- No averaging down.
- No quantity rounding that exceeds risk.
- Zero/invalid quantity rejects with reason.

All risk blocks must be visible in UI, reports and diagnostics.

---

23. REALISTIC F&O COSTS AND NET P&L

Centralize a versioned F&O cost model.

It must include, as applicable:

- brokerage
- exchange transaction charges
- clearing charges if modelled
- statutory levies
- STT
- GST
- SEBI charges
- stamp duty
- spread
- slippage

- exercise/expiry behavior
- partial-exit costs

Do not hardcode a tax/charge rate without:

- effective-from date
- official source reference in code documentation
- unit tests
- configurable version

For each trade calculate:

- gross entry value
- gross exit value
- gross P&L
- every cost component
- total costs
- net P&L
- net return %
- R multiple
- MFE
- MAE
- max unrealized P&L
- drawdown during trade
- holding time
- fill latency

Reconcile API, DB, UI, Telegram and exported report values to the same canonical function.

---

24. F&O PAPER-TRADE LIFECYCLE

Use explicit states, for example:

`SIGNAL_CREATED → WATCHLIST/PENDING_TRIGGER → ELIGIBLE → EXECUTION_ATTEMPTED → OPEN → PARTIALLY_EXITED → CLOSED`

Terminal/non-open states:

- `INFO_ONLY`
- `REJECTED`
- `EXPIRED_UNFILLED`
- `CANCELLED`
- `DATA_BLOCKED`
- `EXECUTION_ERROR`

Persist an event log:

- event type
- timestamp
- actor/source
- prior state
- new state
- price/source
- quantity
- reason
- metadata

Use database transactions and idempotency keys.

---

25. F&O SIGNAL QUALITY, EXPECTANCY AND BACKTESTING

Build setup-wise measurement, not one blended win rate.

Break down by:

- index
- setup
- direction
- regime
- tier
- confidence bucket
- time of day
- weekday
- expiry distance
- VIX/IV bucket
- gap type
- option moneyness
- spread bucket
- market-data quality
- exit reason

Metrics:

- sample size
- win rate
- average win/loss

- expectancy
- profit factor
- median R
- max drawdown
- MFE/MAE
- false breakout rate
- unfilled rate
- rejection reasons
- signal-to-open conversion
- gross versus net
- cost drag
- consecutive losses
- recovery time

Rules:

- No “A+” or “high conviction” label without sufficient validated sample and positive expectancy.
- New/low-sample setups must display insufficient-evidence status.
- Preserve the post-P0 clean baseline rule: review post-fix sample only after at least five trading sessions or 20+ post-P0 signals across NIFTY/BANKNIFTY/SENSEX.
- Label older data `PRE_P0_FIX_DATA`.
- Separate real-chain, modelled and synthetic results.
- Use walk-forward/out-of-sample validation.
- Do not tune thresholds on the same sample used for final evaluation.

---

26. F&O REPORTS AND EXPORTS

Create a professional report system with filterable UI plus CSV/XLSX export.

Mandatory reports:

1. Index-wise summary.
2. Date-wise summary.
3. Trade-wise ledger.
4. Setup-wise performance.
5. Signal funnel:
   - candidates
   - emitted signals
   - tradeable
   - rejected
   - execution attempted
   - opened
   - closed
6. Skip/rejection reasons.
7. Open positions and MTM.
8. Gross versus net P&L.
9. Charges/cost breakdown.
10. MFE/MAE and R analysis.
11. Exit reason analysis.
12. Confidence/tier calibration.
13. Pre-P0 versus post-P0.
14. Data-quality impact.
15. Option-chain shadow-confirmation analysis.
16. Index-level capital/risk usage.
17. Daily, weekly, monthly cumulative equity curve and drawdown.
18. Telegram delivery audit.

Trade-wise export columns must include all plan, execution, lifecycle, P&L, provenance and audit fields. No important field may exist only in a popup.

---

27. F&O UI AND TELEGRAM

F&O cockpit must show:

- market/session/data health
- trusted source and as-of
- regime
- bias
- current setup state
- signal tier
- option contract
- planned entry/SL/targets
- execution status
- live premium/MTM
- warnings
- all skip/block reasons
- active/open/closed lifecycle
- net P&L
- expiry/liquidity context

Telegram alerts:

- Send only owner-authorized alerts.
- Distinguish `INFO_ONLY`, `WATCHLIST_SETUP`, tradeable signal, paper open, partial exit and final close.
- Include source/as-of, setup, contract, entry, SL, targets, size, reason and risk.

- Deduplicate persistently.
- Label sample/test messages explicitly.
- Log delivery outcome without exposing secrets.

PART C — SWING CASH ENGINE

28. SWING OBJECTIVE AND UNIVERSE

Target holding period: approximately 5–20 trading sessions, subject to actual strategy design and evidence.

Audit the active universe:

- NIFTY 500 or configured liquid NSE cash universe
- F&O stocks/high-liquidity names if included
- midcap inclusion rules
- exclusions
- inactive/renamed/suspended securities
- value-traded/liquidity filters

Do not silently scan only a partial universe. Show:

- expected count
- resolved count
- candle-ready count
- successfully scored count
- rejected count
- error count
- cached/partial status
- as-of

---

29. SWING DATA INPUTS

Audit source, freshness, formula and actual use for:

- adjusted daily candles
- weekly candles
- optional intraday trigger candles
- price/volume
- ATR
- moving averages
- RSI
- ADX/DMI
- relative volume
- delivery %
- value traded
- 52-week high/low
- stock relative strength versus benchmark
- stock relative strength versus sector
- sector rank/rotation
- market regime
- breadth
- earnings/results calendar
- corporate actions
- pledging/fundamental fields if used
- gap/event risk
- support/resistance
- breakout base
- demand zone/order block/SMC fields

Every factor shown in UI must state whether it actually changes the score/trade decision or is display-only.

---

30. SWING SCORING ENGINE

Map every current component, expected to include items such as:

- technical score
- SMC score
- volume score
- momentum score
- fundamental score
- risk score
- context score
- relative-strength score
- sector score
- liquidity score

For each:

- raw formula
- input fields
- null behavior
- cap/weight

- normalization
- contribution to final score
- reason text
- tests
- historical usefulness

Verify whether the final score is raw total, normalized percentage or another formula. Ensure server, database and UI use the same score.

Mandatory consistency:

- A high score with `AVOID`/`NO_TRADE` must show the exact overriding reason: poor liquidity, event risk, invalid stop, sector weakness, bad data, excessive gap, poor reward-to-risk, duplicate exposure, etc.
- Recommendation labels and paper execution eligibility must not diverge silently.
- Do not let missing optional fundamentals create a false zero/negative score without a documented policy.

---

31. SWING SETUPS

Audit and make explicit:

- breakout
- breakout retest
- pullback in uptrend
- trend continuation
- relative-strength leader
- sector-leader continuation
- volume contraction pattern
- demand-zone/order-block reversal if genuinely implemented
- mean reversion, if used
- any hidden/legacy setup

For each setup define:

1. Required market regime.
2. Required stock trend.
3. Sector confirmation.
4. Relative-strength rule.
5. Liquidity rule.
6. Base/structure quality.
7. Volume/delivery confirmation.
8. Trigger.
9. Entry zone.
10. Stop.
11. Targets.
12. Time stop.
13. Invalidation before entry.
14. Gap handling.
15. Earnings/corporate-event handling.
16. Minimum reward-to-risk.
17. Recommendation label.
18. Paper execution eligibility.
19. Re-entry/cooldown.
20. Historical expectancy.

---

32. SWING CANDIDATE LIFECYCLE

Implement an explicit lifecycle:

`SCANNED → REJECTED or WATCHLIST → NEAR_TRIGGER → TRIGGERED → STAGED → APPROVAL_REQUIRED/APPROVED_DRY_RUN → OPEN → PARTIALLY_EXITED → CLOSED/INVALIDATED/EXPIRED`

Persist all transitions and reasons.

Existing Telegram staged-order events to preserve and verify:

- `STAGED`
- `APPROVAL_REQUIRED`
- `EXPIRED`
- `REJECTED`
- `APPROVED_DRY_RUN`
- `BLOCKED_BY_RISK`

The staged limit is the intended entry; risk evaluation must use current trusted Kite LTP where that is the verified design. Do not rewrite this price-honesty rule without evidence.

---

33. SWING ENTRY RULES

A Swing paper trade can open only when:

- Data is trusted and sufficiently fresh.
- Symbol is active and resolved.
- Required candle history is complete.

- Market regime permits the setup.
- Stock trend/structure qualifies.
- Sector and benchmark context meet the setup rule.
- Relative strength meets the rule.
- Liquidity/value traded is adequate.
- Trigger occurs; watchlist status alone is not an entry.
- Gap from intended trigger is within an approved limit.
- Results/corporate action/event rule permits entry.
- Stop is valid.
- Reward-to-risk meets configured minimum.
- Position size is non-zero and within risk/heat/cash limits.
- No duplicate/conflicting position.
- Staged-order expiry has not passed.
- Paper execution uses an auditable fill rule.

Entry styles to validate separately:

- Buy-stop breakout above trigger.
- Limit/retest entry.
- Pullback confirmation entry.
- Next-session/open gap behavior.
- Partial fill, if modelled.

Do not use future candle information to create the entry.

---

34. SWING STOP-LOSS RULES

Use setup-specific structure, with ATR sanity.

Possible stop anchors to audit/validate:

- breakout base low
- retest failure
- latest valid swing low
- demand-zone invalidation
- moving-average/structure close
- ATR-based buffer

Rules:

- Stop must be known before trade creation.
- No stop widening after entry.
- Reject trades with impractically tight/wide stops.
- Split-adjusted prices/corporate actions must not create false stop events.
- Record intraday hit versus end-of-day close logic explicitly.
- Gap-through-stop paper fill must use a conservative, documented fill rule.

---

35. SWING TARGET, TRAIL AND EXIT RULES

Audit and define:

- T1 based on R multiple/near resistance.
- T2 based on measured move/structure.
- Optional partial booking.
- Breakeven shift only under an explicit rule.
- ATR/chandelier trail.
- swing-low trail.
- moving-average close trail.
- relative-strength deterioration.
- sector breakdown.
- market-regime deterioration.
- earnings/event exit.
- time exit.
- stagnation exit.
- stop hit.
- manual owner close.

Track:

- planned versus actual exit
- partial quantities
- exit source/time
- gap/slippage
- gross/net P&L
- MFE/MAE
- holding sessions
- exit reason

---

36. SWING POSITION SIZING AND PORTFOLIO RISK

Canonical sizing inputs:
- account equity

- available cash
- entry
- stop
- per-trade risk
- lot/quantity increment
- max position value
- available slots
- current sector exposure
- correlation/overlap
- total portfolio heat
- daily/weekly/monthly drawdown latches
- existing open positions

Rules:

- Quantity based on risk and cash; choose the lower permitted quantity.
- No negative/zero/NaN sizes.
- No hidden leverage for cash Swing.
- Max positions.
- Max sector exposure.
- Max single-stock exposure.
- Portfolio heat cap.
- Drawdown controls.
- New-trade block after breached risk cap.
- Cash reservation and release.
- Reconcile staged, open and closed capital.

Every rejection must show the calculated values and reason.

---

37. SWING PAPER-TRADE P&L AND REPORTS

Calculate from actual simulated equity fills, not scanner CMP snapshots.

Include:

- buy value
- sell value
- brokerage/cost assumptions
- slippage
- gross P&L
- net P&L
- return %
- R multiple
- MFE/MAE
- holding sessions
- target/stop/trail/time exit
- capital deployed
- sector
- setup
- score/recommendation at entry
- score components
- market/sector regime
- data source/as-of

Reports:
- date-wise
- trade-wise
- symbol-wise
- setup-wise
- sector-wise
- score bucket
- recommendation label
- market regime
- open versus closed
- rejection reasons
- staged-order conversion
- win rate/expectancy/profit factor/drawdown
- STRONG_BUY versus BUY/WATCHLIST outcome comparison
- sector leader versus non-leader comparison

Provide CSV/XLSX export.

---

38. SWING BACKTESTING AND VALIDATION

Use adjusted historical candles and avoid survivorship/lookahead bias where feasible.

Validate:

- universe membership by date
- delisted/renamed symbols
- corporate actions
- signal timestamp versus entry timestamp
- next-bar fill rules
- gap behavior

- costs/slippage
- sector/benchmark data availability
- walk-forward periods
- out-of-sample results

Do not treat today’s NIFTY 500 constituents as the historical universe without disclosure.

No score threshold or label may be promoted as high quality without sample-size and expectancy evidence.

---

39. SWING UI AND ALERTS

Swing cockpit must show:

- universe/data coverage
- market regime
- sector leaders/laggards
- ranked candidates
- near-trigger list
- staged/approval queue
- active trades
- rejected candidates
- invalidated setups
- score breakdown
- exact entry/SL/targets
- risk/quantity
- reasons
- source/as-of
- lifecycle/P&L

No empty default screen that hides scanner failures. Empty state must say whether there were no candidates, no data, a failed scan, a partial scan or all candidates were rejected.

PART D — CROSS-ENGINE REPORTING, TESTING AND PRODUCTION PROOF
40. DATABASE AND API CONTRACTS

Audit all trading tables and add only necessary migrations with rollback.

Recommended logical entities:

- market_data_health
- instrument_master
- candle/candle_sync_run
- option_chain_snapshot
- signal
- signal_gate_result
- immutable_signal_plan
- execution_attempt
- paper_trade
- paper_trade_event
- partial_fill
- exit_evaluation
- cost_breakdown
- setup_expectancy
- scheduler_run
- trading_day_state
- risk_state
- alert_delivery
- swing_scan_run
- swing_candidate
- staged_order
- report_run

Requirements:

- IDs and foreign keys.
- Idempotency keys.
- Immutable plan columns/audit.
- Strategy/config version.
- Source/as-of.
- Created/updated/closed timestamps.
- Clear nullable semantics.
- Consistent enums in DB, Zod, OpenAPI, generated client and frontend.
- Indexes for day/index/symbol/status/setup queries.
- Retention/archival plan for high-volume data.
- No destructive migration without backup and rollback.

---

41. DIAGNOSTICS AND OBSERVABILITY

Expose owner-only dashboards/endpoints for:

- Kite session
- websocket/subscription state

- quote freshness
- candle freshness/coverage
- option-chain freshness/coverage
- instrument-master counts
- scheduler runs
- F&O signal funnel
- F&O skip reasons
- paper execution errors
- open trade reconciliation
- Swing scan coverage/errors
- Swing rejection reasons
- Telegram delivery
- database latency/errors
- time drift
- duplicate-prevention state

Provide Prometheus-format `/metrics` if already approved, but do not assume the preview container can run a root NTP daemon. A time-drift check can be application-level; production scraping is external infrastructure.
No secrets/tokens in logs or metrics.

---

42. TEST MATRIX

Unit tests

- IST market calendar.
- Data freshness.
- Source/trust classification.
- Instrument resolution.
- Bias calculations.
- Each F&O setup bullish/bearish/invalidation.
- Signal gates.
- Contract selection.
- Risk sizing.
- Cost model.
- P&L.
- Each Swing score component.
- Swing recommendation mapping.
- Entry/stop/target calculations.
- Corporate-action adjustment.
- API schemas.

Integration tests

- Data → F&O signal → plan → paper open → MTM → close → report.
- INFO_ONLY with no paper trade expected.
- WATCHLIST_SETUP pending trigger.
- Execution rejected with reason.
- Stale Kite data blocks trade.
- Restart with open trade.
- Duplicate signal across replicas.
- NIFTY/BANKNIFTY/SENSEX contract resolution.
- Swing scan → ranked candidate → staged → open → close.
- Partial/failed universe.
- Telegram dedup.
- DB migration/rollback.
- OpenAPI codegen preservation of `execution`.

Regression tests for known bugs

- False market-closed state.
- `contractInstrumentToken` mismatch.
- Execution object stripping.
- Wrong `NOT_CONFIRMED` popup.
- IST scheduler/timestamp errors.
- Signal-plan mutation.
- OI unit confusion.
- Gross/net display mismatch.
- MACD warm-up issue.
- Stale cache.
- 198-row partial scan.
- Candle SQL failure.
- Symbol aliases/renames.
- Zero-candidate-to-open visibility.

Manual production smoke tests

- Home.
- Options.
- Option Chain.
- OI Lab.
- Paper Trading.
- Paper Reports.
- Swing scanner/cockpit.
- Strategies.
- Infra Health.
- Kite/session.

- Telegram alerts.
- Exports.
- Browser console.
- API 500 scan.
- Source/freshness badges.
- Market-open status during live session.
- Open/close lifecycle.

---

43. ACCEPTANCE CRITERIA — F&O

Do not declare F&O production-verified until:

- No false market-closed state during live IST market.
- `/api/options/signals` returns valid 200 responses in normal, no-signal and degraded states.
- `execution` survives DB → API → client → UI.
- INFO_ONLY never shows false execution failure.
- Fresh Kite/trusted live data is mandatory for new trades.
- Contract, expiry, strike, lot and token are verified.
- Signal plan is immutable.
- Every emitted/rejected signal has reasons.
- Paper fill is premium-based and auditable.
- Stop/target/exit events are deterministic.
- Forced exit works after restart.
- Gross/net and all costs reconcile.
- Index/date/trade-wise reports reconcile to DB.
- Telegram alert matches the actual stored trade.
- No duplicate opens.
- Post-P0 sample is correctly separated.
- Remaining shadow/live-sample items are labelled honestly.

---

44. ACCEPTANCE CRITERIA — SWING

Do not declare Swing production-verified until:

- Full intended universe coverage is visible.
- All symbol errors are resolved or explicitly listed.
- Candles and benchmark/sector data are fresh and complete enough.
- Score formulas and labels are consistent across code/API/UI.
- High score plus AVOID always has an overriding reason.
- Candidate lifecycle is persisted.
- Triggered entry is distinct from watchlist.
- Stop/target/trail/time-exit rules are explicit.
- Position sizing reconciles with risk and cash.
- Corporate actions/results risks are handled.
- Staged Telegram lifecycle is accurate and deduplicated.
- Trade-wise P&L and reports reconcile.
- Setup/score/sector outcome analytics exist.
- Backtest and paper results are clearly separated.

---

45. REQUIRED DELIVERABLES

Produce all of the following:

1. `FNO_SWING_BASELINE_AND_CODE_MAP.md`
2. `FNO_INTRADAY_DEEP_AUDIT.md`
3. `SWING_CASH_DEEP_AUDIT.md`
4. `DATA_SOURCE_FRESHNESS_MATRIX.md`
5. `TRADING_LOGIC_IMPLEMENTED_VS_REFERENCE.md`
6. `P0_P1_FIX_PLAN.md`
7. `FIX_LOG.md`
8. `DB_MIGRATION_AND_ROLLBACK_PLAN.md`
9. `FNO_SIGNAL_FUNNEL_AND_SKIP_REASON_REPORT.md`
10. `SWING_SCAN_COVERAGE_AND_REJECTION_REPORT.md`
11. `PAPER_TRADING_RECONCILIATION_REPORT.md`
12. `FNO_PNL_INDEX_DATE_TRADE.xlsx`
13. `SWING_PNL_DATE_TRADE_SYMBOL.xlsx`
14. `TEST_AND_PRODUCTION_EVIDENCE.md`
15. `FINAL_GO_NO_GO_REPORT.md`

The fix log must contain:

| Task ID | Priority | Issue | Root Cause | File/Function | Change | Migration | Tests | Dev Status | Prod Status | Live Sample | Residual Risk |

---

46. PRIORITY ORDER

P0 — correctness and financial truth

- False market-closed bug.
- API/Zod/OpenAPI execution truth.

- Trusted live-data enforcement.
- Kite pre-open readiness/degraded banner.
- Entry-to-exit premium correctness.
- Signal-plan immutability.
- Duplicate/restart safety.
- Cost/net P&L reconciliation.
- IST/time/calendar correctness.
- Skip/execution diagnostics.
- Critical Swing symbol/universe/data errors.

P1 — reliability and decision quality

- Candle warehouse/backfill.
- Full-scan timeout/partial-data honesty.
- Bhavcopy/data-health.
- Option-chain shadow confirmation.
- Contract/liquidity selector.
- F&O candidate-to-open audit.
- Swing score/sector/RS integration proof.
- Setup-wise reports and expectancy.
- Backtest provenance.
- Telegram/report audit.

P2 — validated strategy enhancements

- Partial booking.
- T1/breakeven/runner.
- Advanced trails.
- Hard-gating option-chain fields after shadow review.
- Additional F&O indices.
- New Swing setup types.
- Any threshold optimization.

Do not start P2 before P0/P1 evidence is complete.

---

47. FINAL INSTRUCTIONS

- Be brutally honest.
- Do not use “looks correct” as evidence.
- Do not hide missing live proof behind unit tests.
- Do not change strategy thresholds merely to increase the number of trades.
- Do not loosen risk gates just because valid candidates produce zero opens.
- Do not introduce fake or modelled values into live/paper-trade decision paths.
- Do not use Yahoo for F&O trade decisions.
- Do not activate live broker execution.
- Do not remove existing safety controls.
- Do not make broad refactors without a proven root cause.
- Do not mark production verified without deployment and live sample evidence.
- Clearly separate:
  - code implemented
  - dev verified
  - production deployed
  - production infrastructure verified
  - live-market sample pending
  - statistically validated

The final result must be a source-aware, freshness-aware, explainable and restart-safe F&O Intraday and Swing Cash platform in which every signal, rejection, entry, stop, target, exit, charge and P&L number can be traced back to the exact trusted data and rule that produced it.

APPENDIX B — Daily_Market_Analysis_Prompts (2).md (uploaded by owner, verbatim)

````
# Daily Market Analysis Prompts — Nifty / BankNifty / Sensex F&O

Two ready-to-use prompts for a professional intraday/swing trading routine. Run the Pre-Market prompt before 9:00 AM
IST and the Post-Market prompt after 3:45 PM IST (or after the FII/DII + OI data updates, usually by ~6-7 PM).

---

## 1. PRE-MARKET ANALYSIS PROMPT

```
Act as a professional Indian equity F&O intraday/swing trading analyst.
Give me a complete pre-market briefing for today, [DATE], for Nifty 50,
Bank Nifty, and Sensex. Structure the output exactly as below. Use only
verified/current data — flag anything unavailable rather than guessing.

1. OVERNIGHT GLOBAL CUES
   - US markets closing levels & % change (Dow, S&P 500, Nasdaq) + key drivers
   - Asian markets today (Nikkei, Hang Seng, Shanghai Composite, Kospi, Taiwan)
   - European markets previous close (FTSE, DAX, CAC)
   - US 10-Year Treasury yield, Dollar Index (DXY)
   - Crude oil (Brent & WTI) — level and overnight move
   - Any overnight geopolitical/macro trigger

2. GIFT NIFTY / SGX NIFTY
   - Current GIFT Nifty level vs Nifty previous close
   - Implied gap-up/gap-down in points and %
   - Range GIFT Nifty has traded in overnight (if available)

3. FII / DII ACTIVITY (PREVIOUS SESSION)
   - Cash market: FII net buy/sell, DII net buy/sell (■ cr)
   - F&O: FII net position change in Index Futures (long/short, contracts, ■ cr)
   - FII Index Options — net premium bought/sold (calls vs puts)
   - Net FII long-short ratio in index futures (current %)

4. PARTICIPANT-WISE OPEN INTEREST (DAILY CHANGE)
   - FII, DII, Pro, Client — Index Futures OI (long %, short %, net change vs prior day)
   - Same breakdown for Index Options (calls & puts separately if data allows)
   - Stock Futures OI trend (aggregate long/short build-up)
   - Interpretation: who is adding/unwinding, and what it implies for bias

5. INDIA VIX
   - Current level vs previous close, % change
   - Implied volatility read — complacency or fear building

6. KEY LEVELS — PRICE ACTION + OPTION-CHAIN BASED
   For Nifty, Bank Nifty, and Sensex individually:
   - Previous day High / Low / Close, Open
   - CPR (Central Pivot Range) & standard pivot levels
   - Price-action Support: S1, S2, S3 (swing lows / demand zones)
   - Price-action Resistance: R1/T1, T2, T3 (swing highs / supply zones)
   - Option-chain based support/resistance: highest Put OI strikes (S1/S2/S3)
     and highest Call OI strikes (T1/T2/T3)
   - Max Pain level
   - PCR (OI-based and volume-based)
   - Any major OI concentration shift vs previous day

7. EXPECTED RANGE FOR THE DAY
   - Range implied by ATM straddle premium
   - India VIX-implied 1-day move (VIX/√252 approach)
   - ATR(14)-based expected range
   - Final consolidated range estimate with rationale

8. NEWS & EVENTS
   - Domestic: RBI/government policy, macro data releases due today (CPI, IIP,
     PMI, GST collections, trade data etc.), major corporate earnings/events
   - Global: Fed/ECB commentary, central bank events, major economic releases
     (US jobs, CPI, FOMC etc.), geopolitical developments
   - Sector-specific triggers (banking, IT, auto, pharma, metals, energy)
   - Corporate actions today: results calendar, board meetings, bulk/block deals
9. EXPIRY / ROLLOVER CHECK (if applicable today)
   - Is today a weekly/monthly expiry for Nifty/BankNifty/Sensex?
   - Rollover % and cost-of-carry if near monthly expiry
   - Any unusual open interest buildup at expiry strikes

10. BIAS & TRADE PLAN
    - Overall bias: bullish / bearish / range-bound, with 2-3 line rationale
      combining global cues + OI positioning + option data
    - Key trigger levels for long vs short bias (above X = bullish, below Y = bearish)
    - Opening range strategy note (gap-up/gap-down handling)
    - Risk flags for today (event risk, low liquidity, holiday-adjacent session etc.)

Keep it data-first and concise — numbers and levels over narrative. Where
````
````

data is unavailable or stale, say so explicitly instead of estimating.
```

---

## 2. POST-MARKET ANALYSIS PROMPT

```
Act as a professional Indian equity F&O intraday/swing trading analyst.
Give me a complete post-market wrap-up for today, [DATE], for Nifty 50,
Bank Nifty, and Sensex. Structure the output exactly as below.

1. INDEX PERFORMANCE
   - Closing levels, points & % change for Nifty, Bank Nifty, Sensex,
     Nifty Midcap 100, Nifty Bank vs Nifty Financial Services
   - Day's High/Low/Open/Close, closing basis (near high/low/mid-range)
   - Candle structure interpretation (e.g. bullish engulfing, doji, long
     upper/lower wick) and what it signals for tomorrow

2. MARKET BREADTH
   - Advance/Decline ratio (NSE & BSE)
   - New 52-week highs vs lows
   - % of stocks above key moving averages (20/50/200 DMA) if available
   - Sectoral breadth — which sectors led/lagged

3. FII / DII ACTIVITY (TODAY, PROVISIONAL)
   - Cash market net buy/sell (■ cr) — FII and DII
   - F&O: FII index futures net change (long/short, ■ cr)
   - FII index options — net premium flow (calls vs puts)
   - Net long-short ratio change vs pre-market read

4. PARTICIPANT-WISE OI CHANGE (END OF DAY)
   - FII/DII/Pro/Client — net OI change in index futures & options
   - Long unwinding vs short covering vs fresh long/short buildup —
     explicit classification for each participant category
   - Stock futures aggregate OI trend

5. OPTION CHAIN — END OF DAY CHANGE
   For Nifty, Bank Nifty, Sensex:
   - OI addition/unwinding at each major strike (top 3-5 strikes each side)
   - PCR shift (open vs close)
   - Max Pain shift vs pre-market level
   - New support/resistance zones formed for tomorrow (S1-S3, T1-T3)
   - India VIX close vs open, and what the change implies

6. LEVEL VALIDATION
   - Did pre-market S1/S2/S3 and T1/T2/T3 hold, break, or get retested?
   - CPR/pivot respected or violated — note for tomorrow's playbook
   - VWAP behavior through the session (trend day vs mean-reversion day)

7. SECTOR & STOCK MOVES
   - Top gaining/losing sectors with key drivers
   - Top gainers/losers by % and by value (large caps)
   - Volume shockers / unusual activity
   - Any stock-specific news that moved the index materially

8. NEWS RECAP
   - What actually drove the session (confirm/refute the pre-market thesis)
   - Any surprise data releases, RBI/government commentary, global shocks
   - Corporate results announced post-market with likely next-day impact

9. GLOBAL STATUS CHECK (LIVE AS OF NOW)
   - European markets currently trading — direction
   - US futures (if available at time of writing)
   - GIFT Nifty after-hours trend, if session is open

10. TOMORROW SETUP
    - Key events/data scheduled for tomorrow (domestic + global)
    - Expiry/rollover note if relevant
    - Preliminary bias for tomorrow based on today's close + OI positioning
    - Watch-list levels carried forward

11. TRADE JOURNAL TIE-IN (self-review)
    - Did today's actual price action match the pre-market bias/plan?
    - Where did the plan work / fail, and why (data-driven, not emotional)
    - One-line discipline note for tomorrow (position sizing, no-trade
      triggers, or emotional flags based on today's P&L pattern)

Keep it data-first and concise. Flag any figures that are provisional
(e.g. FII data is usually provisional until the next morning) and any
data that's unavailable rather than estimating.
```

Notes on what's covered beyond your original list

- **India VIX** — implied volatility is the backbone of range prediction and often missed
```

- **FII/DII cash vs F&O split** — cash flows and derivative flows tell different stories and are often conflated
- **DXY, US 10Y yield, crude oil** — macro drivers that move Indian markets via currency and inflation channels
- **Market breadth (A/D ratio, new highs/lows)** — tells you if the index move is broad-based or narrow (index-heavy stock driven)
- **Expiry/rollover data** — critical near monthly/weekly expiries, especially for BankNifty/Sensex where OI can be thin on non-benchmark days
- **Candle/VWAP structure validation in post-market** — closes the loop on whether your pre-market levels actually worked, which feeds directly into your trade journal
- **Trade journal tie-in** — given the emotional-override pattern we've discussed on high-loss days, a one-line self-review each evening (not judgmental, just factual) is worth keeping as a standing habit alongside the data recap

You can drop `[DATE]` and just say "today" when using these live. If you want, I can also build a version of these prompts wired directly into Market Scanner's data feeds (participant OI, option chain, VIX) so it auto-generates this briefing instead of you running it manually each day.

APPENDIX C — PRE_POST_BRIEFING_MASTER_PROMPT.md (authored by Claude, verbatim)

```
# MASTER CODER PROMPT — PRE/POST MARKET BRIEFING ENGINE + TELEGRAM DELIVERY + PRE/POST TAB

**Project:** Market Scanner by Dev — Indian Market Intelligence and Trading Decision Platform
**Feature:** Automated institutional-grade Pre-Market Briefing and Post-Market Wrap for NIFTY 50, BANKNIFTY, SENSEX — rendered in a dedicated **Pre/Post tab** and delivered via **Telegram** (extending the existing pre/post-market Telegram report generators, NOT replacing them).
**Operating mode:** Read-only analytics feature. This feature makes ZERO trading decisions, touches ZERO signal/paper-trade write paths, and places ZERO orders.
**Time standard:** Every timestamp, scheduler, and displayed time uses `Asia/Kolkata` (IST). No exceptions.

---

## 0. NON-NEGOTIABLE CONSTRAINTS (READ FIRST — THESE OVERRIDE EVERYTHING BELOW)

1. **DRIFT-P0 IS ACTIVE.** No merges that trigger `scripts/post-merge.sh`, no manual `drizzle-kit push`, until the Drizzle-drift-reconciliation slice has landed and the owner has confirmed it. Build this feature on a branch. Merging is a separate, owner-approved event.
2. **Data honesty is the product.** Never fabricate, never estimate silently, never present stale as live, never present derived as observed. Every rendered number carries a provenance state: `LIVE` / `EOD` / `STALE` / `UNAVAILABLE`. A section whose source is down renders an explicit UNAVAILABLE block — it does not disappear, and it does not guess.
3. **Schema bright line:** additive only. New columns via `ALTER TABLE ... ADD COLUMN IF NOT EXISTS` applied directly. Any statement that is not literally ADD COLUMN IF NOT EXISTS (including type changes, constraints, defaults) requires owner pre-approval BEFORE execution. **New tables must be declared in the Drizzle TS schema on the same day they are created** — no new runtime-only tables; we are not adding to the drift we are about to reconcile.
4. **No adjacent actions.** Do exactly what each phase scopes. Discovering a bug elsewhere = log it and report; do not fix it inside this feature's phases. Specifically: the corrupted `vix` column in `fno_signal_reasoning` is a SEPARATE ticket — this feature works around it (see §3.5), it does not fix it.
5. **No market-hours deploys/restarts.** All restarts and flag flips happen after 15:30 IST or before 09:00 IST.
6. **No new third-party data providers, scraping workarounds, or paid APIs without owner approval.** Where a section needs a source the platform doesn't have (bhavcopy from prod IP, global macro, news), the deliverable is a PROPOSAL with options and costs — not an implementation.
7. **Acceptance evidence is literal:** pasted test pass lines (`pnpm --filter @workspace/api-server run test` in Replit Shell, `--pool=threads`), typecheck exit 0, real screenshots, real Telegram messages delivered to the owner chat, and fail-closed tests proving every gated section renders UNAVAILABLE when its source is down.
8. **Preserve existing strengths:** the existing pre-market and post-market Telegram reports keep working throughout. This feature extends their generators; it must never break or silently replace them until the owner explicitly retires the old format.

---

## 1. FEATURE OVERVIEW

Two generated artifacts per trading day, each with three delivery surfaces:

| Artifact | Generated (IST) | Surfaces |
|---|---|---|
| **Pre-Market Briefing** | 08:15 (configurable), ready before 09:00 | Pre/Post tab, Telegram, JSON API |
| **Post-Market Wrap** | Two-stage: **Quick Wrap** at 15:45; **Full Wrap** at 18:45 after provisional FII/DII + participant data windows | Pre/Post tab, Telegram, JSON API |

Both artifacts are **persisted** (see §4 schema) so that: (a) the Post-Market Wrap can validate the same day's pre-market levels (§3, post §6), (b) history is browsable in the tab, (c) the trade-journal tie-in accumulates over time.

**Section-gated architecture (core design):** every section of both briefings is an independent module with:
- a declared data source and provenance state,
- a `sectionStatus`: `ACTIVE` | `PARTIAL` | `GATED` (source not yet trusted/available on this platform),
- honest rendering in all three states.

GATED sections ship in v1 as visible, labeled placeholders ("Participant-wise OI — UNAVAILABLE: NSE participant data not yet accessible from production. Tracking: <ticket>"). They unlock individually, each unlock being its own owner-approved mini-slice when the upstream source lands. **The briefing must be useful and honest on day one with only the Kite-derived sections active.**

---

## 2. PHASE 0 — READ-ONLY DATA-SOURCE AUDIT (NO CODE; FIRST DELIVERABLE)

Before any code, produce a **Data Availability Matrix**: for every section in §3, verify with evidence (actual API responses, actual file checks, actual Kite quote pulls in a scratch script) what the platform can truthfully source TODAY. Expected findings to confirm or refute — do not assume, verify:
| # | Data need | Expected status | Verify |
|---|---|---|---|
| A | Index/candle data, EMA/RSI/VWAP/ATR, prev OHLC | AVAILABLE (Kite, trusted layer) | Pull sample |
| B | Option chain (OI by strike, premiums, PCR, Max Pain inputs) | AVAILABLE (trusted option-chain migration) | Pull sample for all 3 indices |
| C | India VIX **level** | AVAILABLE via direct Kite quote (`NSE:INDIA VIX`) — must NOT be read from the corrupted `fno_signal_reasoning.vix` column | Pull live quote, sanity 8–35 range |
| D | Instrument master expiries (weekly/monthly per index) | AVAILABLE | Query |
| E | ATM straddle premium (expected-range input) | AVAILABLE (derived from B) | Compute sample |
| F | NSE participant-wise OI / FII-DII F&O breakdown (bhavcopy & participant files) | GATED — NSE blocked from prod cloud IP (known Stage 2 item) | Attempt fetch, capture the actual failure |
```

G	FII/DII **cash** provisional (exchange EOD publications)	VERIFY — may be GATED same as F	Attempt
H	GIFT Nifty	GATED — not in Kite; needs external source	Confirm absence
I	Global markets (US/Asia/Europe closes), US 10Y, DXY, crude	GATED/UNTRUSTED — Yahoo provider flaky; US 10Y parsing had a known defect (0.45 vs 4.5%)	Document current provider state; do NOT trust it silently
J	News/events/results calendar	GATED — no trusted source	Confirm
K	Market breadth (full-exchange A/D, 52w highs/lows)	PARTIAL — NIFTY 500 universe via Kite is feasible; full-exchange breadth needs F	Test a NIFTY 500 quote sweep for rate-limit feasibility
L	Sector indices performance	VERIFY — Kite has NSE sector indices quotes	Pull sample

Checkpoint 0: paste the completed matrix with raw evidence per row + the resulting v1 section list (ACTIVE/PARTIAL/GATED per section). Owner signs off the matrix before Phase 1. Any section the owner expected ACTIVE that audits as GATED gets flagged loudly here, not discovered in Phase 3.

---

3. SECTION SPECIFICATIONS

Sections marked **[K]** are Kite-derivable and expected ACTIVE in v1. **[G]** = expected GATED pending upstream. **[P]** = PARTIAL. Phase 0 may reclassify.

3A. PRE-MARKET BRIEFING (generated 08:15 IST)

PRE-1. Overnight Global Cues [G] — US closes (Dow/S&P/Nasdaq) + drivers, Asia live (Nikkei/HangSeng/Shanghai/Kospi/Taiwan), Europe prev close (FTSE/DAX/CAC), US 10Y, DXY, Brent & WTI, overnight macro/geopolitical trigger. *Gated until a trusted macro provider is approved. The US-10Y unit-parsing defect must be fixed and regression-tested as part of that provider slice — a yield displayed at 10x/0.1x scale in a trading briefing is a P0-class error.*

PRE-2. GIFT Nifty [G] — level vs NIFTY prev close, implied gap (points/%), overnight range. Gated: propose source options in Phase 0.

PRE-3. FII/DII Activity (previous session) [G] — cash net (■ cr) FII/DII; F&O index-futures net change; index-options net premium flow; FII long-short ratio %. Gated on F/G from the matrix.

PRE-4. Participant-wise OI daily change [G] — FII/DII/Pro/Client index futures long%/short%/net change; index options CE/PE; stock futures aggregate; one-line interpretation per participant (fresh long / fresh short / long unwind / short cover — classified by price+OI logic, formula documented). Gated on bhavcopy access.

PRE-5. India VIX [K] — level (direct Kite quote at generation time), vs prev close, % change, one-line volatility read (thresholds config-driven, not hardcoded). **Hard rule: source = live Kite quote only. Never the `fno_signal_reasoning.vix` column. Sanity gate: reject and mark STALE/UNAVAILABLE if outside 5–60.**

PRE-6. Key Levels [K] — per index: prev day O/H/L/C; CPR + standard pivots; price-action S1–S3 / R1–R3 (swing/structure derived — document the exact algorithm, no hand-waving); option-chain S/R: top-3 Put-OI strikes and top-3 Call-OI strikes; Max Pain; PCR (OI and volume); major OI concentration shift vs prior day (needs stored prior-day chain snapshot — see §4). Every level tagged with its derivation.

PRE-7. Expected Range [P] — (a) ATM straddle-implied range [K]; (b) VIX-implied 1-day move: `spot × VIX/100/√252` [K, uses PRE-5's trusted VIX]; (c) ATR(14)-based [K]; (d) consolidated estimate with stated weighting. If any input is UNAVAILABLE, the consolidated line lists which inputs it actually used.

PRE-8. News & Events [G] — domestic macro calendar, global events, sector triggers, results calendar. Gated. Interim option (owner decision in Phase 0): a manual "owner notes" input on the tab that flows into the briefing, labeled MANUAL.

PRE-9. Expiry/Rollover Check [P] — is today weekly/monthly expiry per index [K, instrument master]; rollover % + cost-of-carry near monthly expiry [G, needs futures OI history]; unusual expiry-strike OI buildup [K via chain snapshots].
PRE-10. Bias & Trade Plan [K-derived] — bias (bullish/bearish/range) with 2–3 line rationale that **cites only ACTIVE sections' data**; trigger levels (above X bullish / below Y bearish) from PRE-6; opening-range note (gap handling: gap size from prev close vs first trusted quote — labeled GIFT-unavailable if PRE-2 gated); risk flags (expiry day, event day if PRE-8 active, holiday-adjacent). **The rationale must list its inputs. A bias computed from partial data says so: "Based on price structure + option chain + VIX only (global cues unavailable)."** This section reuses/aligns with the platform's canonical bias logic where applicable — do not create a third bias definition; if the canonical bias function (P1.2 scope) doesn't exist yet, document that this section computes a DISPLAY-ONLY bias with its own labeled formula, to be reconciled when P1.2 lands.

3B. POST-MARKET WRAP

Quick Wrap (15:45 IST) = sections POST-1, POST-2(partial), POST-5, POST-6, POST-7 with EOD-provisional labels. **Full Wrap (18:45 IST)** = all sections, superseding the quick wrap in the tab (both persisted).

POST-1. Index Performance [K] — close, points/% for NIFTY, BANKNIFTY, SENSEX, Midcap 100 (+ Bank vs FinNifty if quotes available per Phase 0); O/H/L/C; close position within range (near-high/near-low/mid, formula stated); candle-structure classification (rule-based classifier from a documented pattern table — engulfing, doji, hammer, long-wick etc. — with the OHLC math shown; never an unexplained label).

POST-2. Market Breadth [P] — v1: NIFTY 500-universe A/D ratio, new 52w highs/lows within universe, % above 20/50/200 DMA (from candle warehouse — verify coverage in Phase 0), sector-level breadth. Full-exchange breadth unlocks with bhavcopy. Label clearly: "Breadth universe: NIFTY 500".

POST-3. FII/DII Activity today, provisional [G] — as PRE-3, marked PROVISIONAL.

POST-4. Participant-wise OI change EOD [G] — as PRE-4 + explicit per-participant classification (long unwind / short cover / fresh long / fresh short).

POST-5. Option Chain EOD Change [K] — per index: OI added/unwound at top 3–5 strikes each side (from open-snapshot vs close-snapshot — see §4 snapshots); PCR shift open→close; Max Pain shift vs pre-market; tomorrow's S/R zones from closing chain; India VIX open vs close (direct quotes captured by the snapshot job) and implication.

POST-6. Level Validation [K] — the loop-closer: load the SAME DAY's persisted pre-market levels; for each S1–S3/R1–R3/CPR: HELD / BROKE / RETESTED / UNTESTED against the session's actual candles (algorithm documented: touch tolerance in points/%, close-through rules); VWAP behavior classification (trend day vs mean-reversion day, rule stated). If no pre-market briefing exists for today, say exactly that.

POST-7. Sector & Stock Moves [P] — top gaining/losing sectors [per Phase 0 row L]; top large-cap gainers/losers % (universe labeled); volume shockers (volume vs N-day average, threshold config-driven); index-moving stock news [G — omit or MANUAL].

POST-8. News Recap [G] — what drove the session vs pre-market thesis. Interim: MANUAL owner notes.

POST-9. Global Status Live [G] — Europe live direction, US futures, GIFT after-hours.

POST-10. Tomorrow Setup [K-derived] — tomorrow's expiry/event flags [K]; carried-forward watch levels (closing chain + structure); preliminary bias for tomorrow with cited inputs, same honesty rule as PRE-10.

POST-11. Trade Journal Tie-in [K + MANUAL] — auto part: today's platform activity summary (signals emitted by class, paper trades opened/closed + net P&L when the paper engine goes live — until then: "Paper engine: no live trades — funnel summary only" with the day's funnel counts from `fno_signal_reasoning`); manual part: a one-line owner self-review input on the tab ("Did the plan match? One discipline note.") persisted per day, shown in the wrap, and browsable as a journal history view. **The manual journal input is deliberately friction-free: one text field, one save button.**

---

4. SCHEMA (PROPOSAL-FIRST — checkpoint before any DDL)

Propose in Phase 1, apply only after owner sign-off. Expected shape (adjust to audit findings):

- **`daily_briefings`** — `id`, `briefing_date` (IST date), `kind` (`PRE` | `POST_QUICK` | `POST_FULL`), `generated_at`, `payload_json` (full structured briefing: every section with its data, provenance state, and per-number source tags), `section_status_json` (ACTIVE/PARTIAL/GATED/FAILED per section), `telegram_delivery_status`, `telegram_message_ids_json`, `generator_version`. Unique on (`briefing_date`,`kind`).
- **`option_chain_snapshots`** — `id`, `snapshot_date`, `index_symbol`, `snapshot_kind` (`OPEN` ~09:20 | `CLOSE` ~15:25 | `PREV_EOD`), `captured_at`, `chain_json` (strikes/OI/premiums), `vix_quote`, `spot`, `data_quality`. Powers PRE-6 OI-shift, POST-5 deltas, PRE-9 expiry buildup. Retention policy proposed (e.g. 90 days) — owner decides.
- **`owner_journal_entries`** — `id`, `entry_date`, `note_text`, `created_at`, `updated_at`. One row per day, upsert.
- All three tables: created via approved DDL AND declared in Drizzle TS schema in the same PR. Zero new drift.

No changes to any existing signal/paper/trading table. Read-only access to `fno_signal_reasoning` (POST-11 funnel counts) and trusted market-data services.

---

5. SCHEDULER & FAILURE BEHAVIOR

- Jobs (all IST, config-driven times): `PRE_BRIEFING@08:15`, `CHAIN_SNAPSHOT_OPEN@09:20`, `POST_QUICK@15:45`, `CHAIN_SNAPSHOT_CLOSE@15:25`, `POST_FULL@18:45`. Skip on NSE holidays/weekends using the platform's existing market-calendar service (the same trusted source as the marketStatus work — no new calendar logic).
- Reuse the platform's existing scheduler patterns including dedup guards (scheduler dedup is a known P1.1 concern — do not add a new unguarded scheduler class).
- **Failure = loud + partial, never silent + fake.** If generation fails wholesale: persist a FAILED row, send a one-line owner Telegram alert ("Pre-market briefing generation FAILED: <reason>"). If individual sections fail: render them as FAILED/UNAVAILABLE with reason, deliver the rest.
- Manual "Regenerate now" button on the tab (owner-only), which also respects provenance honestly (a 14:00 regeneration of the pre-market briefing labels itself as regenerated-at 14:00).

---

6. TELEGRAM TEMPLATES (v1 — render only ACTIVE/PARTIAL sections; GATED sections collapse to one footer line)

Formatting rules: Telegram HTML parse mode; hard 4,096-char limit per message → deliver as numbered parts ("Part 1/3"); provenance chips: ■ LIVE · ■ EOD · ■ STALE/PROVISIONAL · ■ UNAVAILABLE · ✍■ MANUAL; numbers right-padded alignment where feasible; no emoji noise beyond chips and section markers; every message footer: `Generated <HH:MM> IST · Market Scanner by Dev · <generator_version>`.

6A. PRE-MARKET TEMPLATE (v1, Kite-active sections)

```
■ PRE-MARKET BRIEFING — {DD Mon YYYY} ({Day})
Part {n}/{N} · Generated {HH:MM} IST

■■ INDIA VIX ■
{vix_level} ({vix_chg:+.2f} / {vix_chg_pct:+.2f}%) vs prev close
Read: {one_line_vol_read}

■■ KEY LEVELS — NIFTY ■
Prev: O {o} · H {h} · L {l} · C {c}
CPR: {bc} – {pivot} – {tc}  ({narrow|wide})
PA Support: S1 {s1} · S2 {s2} · S3 {s3}
PA Resistance: R1 {r1} · R2 {r2} · R3 {r3}
OI Support (PE): {pe1} · {pe2} · {pe3}
OI Resistance (CE): {ce1} · {ce2} · {ce3}
Max Pain: {mp} · PCR(OI): {pcr_oi} · PCR(Vol): {pcr_vol}
OI shift vs y'day: {one_line_shift_or_"first snapshot — no prior"}
■■ KEY LEVELS — BANKNIFTY ■
```

```
{same block}

■■ KEY LEVELS — SENSEX ■
{same block}

■■ EXPECTED RANGE (today) ■
Straddle-implied: {index}: ±{pts} ({lo}–{hi})
VIX-implied 1d: ±{pts_vix}
ATR(14): ±{pts_atr}
Consolidated: {lo*}–{hi*}  (inputs: {list})

■■ EXPIRY CHECK ■
{None today | NIFTY weekly expiry TODAY | ...}
{expiry-strike OI note if applicable}

■■ BIAS & PLAN (display-only, not a signal)
Bias: {BULLISH|BEARISH|RANGE} — {rationale, 2 lines, cited inputs}
Long trigger: above {x} · Short trigger: below {y}
Opening note: {gap handling line}
Risk flags: {flags}
■■ Inputs used: {active sections}. Not used (unavailable): {gated list}.

■■ UNAVAILABLE THIS BUILD ■
Global cues · GIFT Nifty · FII/DII · Participant OI · News
(unlock tracked in platform roadmap)

Generated {HH:MM} IST · Market Scanner by Dev · {version}
```

6B. POST-MARKET TEMPLATE (Full Wrap v1)

```
■ POST-MARKET WRAP — {DD Mon YYYY} ({Day})
Part {n}/{N} · Generated {HH:MM} IST

■■ INDEX PERFORMANCE ■
NIFTY     {close}  {chg:+}  ({pct:+.2f}%)
BANKNIFTY {close}  {chg:+}  ({pct:+.2f}%)
SENSEX    {close}  {chg:+}  ({pct:+.2f}%)
MIDCAP100 {close}  {chg:+}  ({pct:+.2f}%)
NIFTY: O {o} H {h} L {l} — closed {near high|near low|mid-range}
Candle: {classification} → {one_line_implication}
{repeat candle line for BN/SENSEX}

■■ BREADTH (NIFTY 500 universe) ■
A/D: {adv}/{dec} ({ratio}) · New 52w H/L: {nh}/{nl}
Above 20/50/200 DMA: {p20}% / {p50}% / {p200}%
Led: {sectors} · Lagged: {sectors}

■■ OPTION CHAIN — EOD CHANGE ■
NIFTY  PCR {pcr_open}→{pcr_close} · MaxPain {mp_pre}→{mp_close}
OI added: {ce_strikes} CE · {pe_strikes} PE
OI unwound: {strikes}
Tomorrow zones: S {s_zones} · R {r_zones}
{repeat for BANKNIFTY, SENSEX}
India VIX: {open}→{close} ({chg_pct:+.1f}%) — {implication}

■■ LEVEL VALIDATION (vs today's pre-market) ■
NIFTY: S1 {HELD|BROKE|RETESTED|UNTESTED} · R1 {...} · CPR {respected|violated}
{repeat per index}
Session type: {trend day|mean-reversion day} (VWAP: {behavior})
Playbook note: {one line for tomorrow}

■■ SECTOR & STOCK MOVES ■
Top sectors: {gainers} / {losers}
Large-cap movers: {names+%}
Volume shockers: {names} (vs {n}d avg)

■■ TOMORROW SETUP
Events: {expiry/known flags} · Watch: {levels}
Preliminary bias: {bias} — {cited inputs}

■■ JOURNAL ■
Platform: {n} signals emitted ({class breakdown}) · Paper: {trades or "no live trades — funnel only"}
Owner note ✍■: {note_text | "— (add today's note in the Pre/Post tab)"}

■■ UNAVAILABLE THIS BUILD ■
FII/DII · Participant OI · News recap · Global live
■ PROVISIONAL: any figure marked ■ may be revised by exchanges.

Generated {HH:MM} IST · Market Scanner by Dev · {version}
```

Templates live in code as versioned template functions with unit tests asserting: chip presence on every numeric block, part-splitting under 4,096 chars, GATED collapse behavior, and the PRE-10 inputs-used disclosure line always present.

---

7. PRE/POST TAB (frontend)
- Dark terminal aesthetic consistent with the NSE option-chain redesign; provenance chips (LIVE/EOD/STALE/UNAVAILABLE/MANUAL) as the visual signature, same chip system as the rest of the platform.
- Layout: date selector (default today) → toggle PRE / POST(QUICK/FULL) → sections as collapsible cards in the §3 order → GATED sections render as visible muted cards with the unlock reason (never hidden — the owner should always see what the briefing is NOT telling him).
- Journal card: today's one-line input + save; history list (last 30 entries) below.
- Owner-only controls: "Regenerate now", "Resend to Telegram".
- Read-only against `daily_briefings` payloads — the tab renders persisted artifacts; it does not recompute.
- History browsing: previous days' briefings selectable; POST-6 validation results visible per day (this becomes the playbook-accuracy record over time).

---

8. DELIVERY PLAN — PHASED, CHECKPOINTED

Phase	Scope	Checkpoint evidence
0	Data Availability Matrix (read-only)	Matrix + raw evidence per row; owner signs section classification
1	Schema proposal → approved DDL + Drizzle declarations; section framework + provenance/gating engine; generator skeleton persisting a briefing with all sections GATED	Tests + typecheck; a persisted skeleton briefing row; fail-closed tests
2	ACTIVE sections: PRE-5/6/7/9/10, POST-1/2/5/6/7/10/11(auto part); chain-snapshot jobs	Tests incl. per-section fail-closed (source down → UNAVAILABLE render); sample generated briefing JSONs from live data pasted
3	Telegram rendering + delivery (extend existing report infra), scheduler jobs, failure alerts, part-splitting	Real messages delivered to owner chat (screenshots); template unit tests; holiday-skip test; dedup test
4	Pre/Post tab UI + journal input + owner controls	Screenshots vs live data; journal round-trip test
5+	Individual unlock slices (bhavcopy/participant OI, macro provider incl. US-10Y fix, GIFT, news) — each begins with a PROPOSAL, its own approval, its own acceptance	Per-slice

One phase at a time. Checkpoint → owner review → next phase. No phase starts early because the previous one "basically passed."

---

9. EXPLICIT DO-NOT LIST

- Do NOT merge to main / trigger post-merge.sh (DRIFT-P0).
- Do NOT read VIX from `fno_signal_reasoning.vix`. Do NOT fix that column here.
- Do NOT touch signal emission, paper trading, gates, thresholds, or any write path of the trading engine.
- Do NOT add data providers, scrapers, or NSE-IP workarounds without an approved proposal.
- Do NOT hide a section because its data is missing — render the absence.
- Do NOT let the display-only bias in PRE-10/POST-10 be consumed by any trading logic. It is annotated `DISPLAY_ONLY` in the payload.
- Do NOT modify or disable the existing Telegram reports.
- Do NOT create any table without its same-day Drizzle declaration.

---

10. DEFINITION OF DONE (v1)

1. Phase 0 matrix signed off; Phases 1–4 checkpointed with literal evidence.
2. For 3 consecutive trading days: pre-market briefing delivered before 09:00 IST and full wrap by 19:00 IST, to both tab and Telegram, with zero fabricated fields — verified by spot-audit comparing rendered numbers against raw Kite pulls.
3. Every GATED section visibly labeled with its unlock ticket.
4. Level-validation loop demonstrated end-to-end on a real day: pre-market levels → session → POST-6 verdicts.
5. Full test suite green (literal lines), typecheck exit 0, no regressions in existing Telegram reports.
6. Owner has entered at least one journal note and it renders in that evening's wrap.

APPENDIX D — FNO_COMPLETION_MISSION.md (authored by Claude, verbatim)

```
# MISSION DIRECTIVE — F&O MONEY-PATH COMPLETION

**Objective:** The platform's first honest, fully-instrumented paper trade — signal emitted by a real production writer, executed with real option premiums, exited by a defined rule, P&L computed net of real NSE costs — followed by a continuous 20+ signal evaluation sample across NIFTY / BANKNIFTY / SENSEX.
**Target:** First real trade within ~10 trading sessions of mission start. Evaluation sample within ~5 sessions after that.
**This is one mission, not six slices.** Each phase below checkpoints ONCE at its end. Sub-steps inside a phase are pre-authorized as scoped here. Everything not on this mission's critical path is frozen until the mission completes.

**Why this exists:** Six months of build has produced a truthful platform and zero trades. The 2026-07-17 session is the proof case: the analysis layer read the day correctly (bullish OI positioning on all three indices), the market delivered +1.1% to +1.7% trend moves, and the system emitted only INFO_ONLY baseline broadcasts whose triggers expired 30–90 minutes before price crossed every one of them. Nothing was wrong with the market read. The tradeable lane does not exist, and the trigger lifecycle is mis-designed. This mission builds the lane and fixes the lifecycle. Nothing else.

---

## STANDING BRIGHT LINES (unchanged, condensed)
1. Schema: `ADD COLUMN IF NOT EXISTS` freely; anything else needs owner pre-approval. New tables get same-day Drizzle declarations.
2. No merges / no Publish until Phase M2 lands (DRIFT-P0).
3. No deploys/restarts during market hours (09:00–15:30 IST).
4. No threshold, veto, clamp, or gate value changes EXCEPT where this mission explicitly scopes them with owner decision recorded.
5. Acceptance evidence is literal: pasted test lines (`--pool=threads`, Replit/Shell), typecheck exit 0, raw query output, screenshots.
6. No adjacent actions. Defects found off-path: log, report, continue.
7. Paper trading only. No live order path work.

---

## PHASE M0 — CLOSE THE LEDGER + BUILD THE EVIDENCE FILE (tonight + weekend)
Scope:
- Finish the existing post-close docket: Items 2–4 (column-width invariant test; paper_trade_eq.quantity diagnosis→proposal; persistence-audit note), Friday probes (Rows F/G/K), and the 9-section acceptance query with the 10:50–11:43 gap annotation. Combined Fri+Mon sample closes P0.4 Step 2.
- Add one read-only capture: **the 2026-07-17 case study** — persist to `/app/memory/forensics/`: (a) all three baseline plan snapshots (triggers, expiry timestamps, MFE/MAE); (b) actual session path: the exact time price crossed each plan's trigger vs. when the plan expired; (c) regime labels vs realized session character (RANGING label on a trend day); (d) timestamps of all 12 "MARKET CLOSED" suppression events — if ANY fired during 09:15–15:30 IST, flag as live P0.1 evidence.
- Deliver the **TRIGGER-GEOMETRY & LIFECYCLE OPTIONS MEMO** (owner decision input, ≤2 pages): using the case study + the earlier 6/6 STALE_TRIGGER evidence, lay out options with pros/cons and the specific numbers from real days for: (1) entry placement model — displaced breakout entry vs pullback/retest entry vs regime-conditional; (2) trigger lifecycle — fixed time-expiry vs re-arm-on-retest vs rolling revalidation; (3) staleness window length; (4) regime gating rules (what may fire in RANGING vs TRENDING vs EXPIRY_DAY). NO recommendations disguised as defaults — options with evidence, owner decides.
Checkpoint M0: docket evidence + acceptance results + case study + memo.

## PHASE M1 — P0.1 + P0.2 (financial truth on the signal surface) — ~2 sessions
Scope (per the master fix doc §4.1 + schema-drift section):
- P0.1: only `marketStatus.marketOpen === false` from a current trusted response renders market-closed. Missing/stale/loading/error → neutral degraded state. Remove deprecated closed-by-default fallbacks. Tests: 09:14 / 09:15 / 12:00 / 15:29 / 15:30 / holiday / special session / missing response / stale cache. If M0 found in-session suppressions, those exact timestamps become regression cases.
- P0.2: `/api/options/signals` returns HTTP 200 on degraded states with `marketStatus` / `setupState` / `dataQuality` / `warnings` / `blockingReasons`; execution fields preserved end-to-end; `contractInstrumentToken` harmonised across DB/Zod/OpenAPI/TS/tests IN THE CODE LAYER (no prod column renames).
Checkpoint M1: tests + typecheck + before/after API samples.

## PHASE M2 — DRIFT RECONCILIATION (unblocks merge/deploy) — ~1–2 sessions
Scope: origin-story inventory of all drift items re-verified against post-recovery reality; TS declarations matching live DB byte-for-byte; `drizzle-kit push` diff = ZERO pending; deployment-env audit (REASONING_WRITER_V2_ENABLED and every writer-read env var present in the Publish deployment config). Closing sub-step: ENV-ISOLATION — provision `nsescanner_test`, add `.env.test`, retire the `!includes("dummy")` idiom in the 17 test files. Owner-side password rotation coordinates here.
Checkpoint M2: empty diff pasted + env audit + first successful merge/Publish. DRIFT-P0 lifts.
## PHASE M3 — PAPER_WRITER DISCIPLINE (the trade path becomes trustworthy) — ~2 sessions
Scope (as previously ruled): `transitionExecutionStatus(fingerprint, from[], to, writerId)` — compare-and-set at SQL level, permission matrix + legal transitions enforced inside it, all four writer sites converted, no app-level read-then-write on execution_status anywhere; TRIGGERED_CLOSED stamp on the paper close path; DB-read validation against closed unions with warn-logging (fingerprint + raw value + writerId); fingerprint widening to varchar(64) full-hash for new writes + comment fix. **VIX rider:** add `VixSnapshot.level`, populate in loadVixSnapshot, fix optionSignals.ts:3191 to write level not intradayPct, 5–80 sanity gate, explicit-NULL + data_quality annotation for VIX-unavailable; denominator = post-cutover rows.
Checkpoint M3: tests (including race-simulation tests for the compare-and-set) + typecheck.

## PHASE M4 — OWNER DECISION GATE (no code)
Owner selects, from the M0 memo: entry model, trigger lifecycle, staleness window, regime gating rules. Decisions recorded in PRD verbatim. This gate is scheduled DURING M1–M3 build time so it never blocks the critical path — memo
```

lands at M0, decision due before M5 starts.

PHASE M5 — P1.2: THE REAL EMITTER + FIRST TRADE — ~3–4 sessions
Scope:
- Build the production TREND_CONTINUATION writer to the master doc §15 contract (regime eligibility, trigger transition not state, confirmation, volume, bias alignment, HTF veto, invalidation, cooldown, dedup, time-of-day, expiry handling, confidence formula, minimum data quality, tier, skip reasons) — implementing the OWNER'S M4 decisions for entry/lifecycle/regime, not defaults.
- Canonical bias function: ONE path consumed by cards, signals, paper, alerts, reports.
- Regime classifier validation against the M0 case study + trailing sessions; regime-compatible gating (no MR in strong trend, trend rules per owner decision).
- Wire emission → contract selection → sizing → paper execution through the existing (now disciplined) paper path: entry from option tick/ask-mid hierarchy (never spot), dual stop (spot invalidation + premium stop), T1/T2 exits + the mandatory exit set already implemented, all writes through transitionExecutionStatus, full funnel rows with values_tested_json populated.
- One additional setup (VWAP_RECLAIM or EMA_PULLBACK, owner picks at M4) MAY follow in the same phase ONLY after TREND_CONTINUATION's first live signals verify clean end-to-end.
Checkpoint M5: the first real paper trade's COMPLETE trace pasted — funnel rows from candidate to open, option-premium entry, lifecycle transitions, exit, net P&L with cost model — plus full suite green. Sites A/B/C acceptance (open since Step 2) closes here.

PHASE M6 — EVALUATION SAMPLE — ~5 sessions, passive + daily spot-checks
Scope: run continuously across all three indices. Daily read-only verification: funnel sums, zero fabricated fields, exits firing per rule. NO tuning during the sample window regardless of P&L — the sample's integrity is the product.
At 20+ signals / 5+ sessions: deliver the evaluation report (per-setup, per-index, per-regime expectancy after costs; funnel conversion; skip-reason distribution; trigger-hit vs stale rates vs the old baseline).
Checkpoint M6: the report. Owner then makes the first evidence-based tuning decisions — which is the moment the 6–12 month out-of-sample clock officially starts.

EXPLICITLY DEFERRED UNTIL MISSION COMPLETE
/audit panel (P0.4 Step 3) · P1.3 exit-price provenance (EXCEPTION: if M5's exit path would write an ambiguous unit, implement the unit-label column then, minimally) · BUG-53/54 signal cards · Briefing engine Phases 1–4 · Row J/J' · Phases 7/8/9 · all backlog items not named above.

FAILURE HANDLING DURING MISSION
Live-market incidents: the emergency lane from 2026-07-17 applies (diagnose → minimal pre-approved fix → evidence → resume). Off-path defects: log and continue. If any phase slips >2 sessions past estimate: stop, report the specific blocker, re-plan with owner — do not silently extend.

DEFINITION OF MISSION SUCCESS
A trade the owner can trust: every number on its trace real, labeled, and reproducible — followed by a sample large enough to start judging the strategy instead of the plumbing. Not "the system is perfect." The system MEASURABLE. Perfection comes from the measurement loop this mission switches on.